

Limited-Memory Renewal Contracts: Exact Eligibility Pooling and Certified Joint Design

Anonymous manuscript for Operations Research

Abstract. A renewal provider must choose a limited command alphabet and allocate an accepted obligation across heterogeneous histories. We develop exact joint optimization for a finite catalog with opening charges, individual participation caps, and realization ceilings. A terminal-first allocation theorem pools histories that admit the same catalog symbols while retaining their different caps and service costs. The resulting correction depends only on the last eligible selected symbol and enters an ordered path recurrence. With common eligibility and quadratic primitives, one polynomial-time call solves the original joint problem, even at interior promises and with arbitrarily many distinct histories. With several eligibility classes, an exact two-sided price oracle provides globally valid bounds over their means. Certified subdivision and an additive approximation scheme depend exponentially on the number of classes rather than on the history count or cost-dispersion bins. Independent primal-dual and coverage checks validate original-space policies and bounds. A declared synthetic study includes 1,024-history exact solutions, paired comparisons, numerical mixed-integer brackets, and all unresolved multi-class cases. The method retains the original command budget and all individual contractual constraints; earlier continuous-design and harmonic-coarsening results remain available under their original assumptions.

Keywords: renewal contracts; dynamic programming; global optimization.

Subject classifications: Dynamic programming: service design; Programming: global optimization.

Area of review: Optimization.

Submission category: Lengthy manuscript.

September 25, 2026

1 Introduction

A renewal provider may distinguish many service histories while its terminal gateway accepts only a small menu of commands. The provider must choose that menu and allocate an accepted obligation across histories. Participation caps restrict expected service, realization ceilings restrict every positive-probability delivery, and opening a command may incur a fixed charge. The menu and the service targets are therefore joint decisions. A fast menu optimizer for prescribed targets does not by itself solve the operational problem.

We show that heterogeneous histories need not create heterogeneous search coordinates. Histories that admit the same catalog commands can be pooled exactly, even when every participation cap and shortfall cost differs. The reason is a terminal-first allocation rule. Within an eligibility class, intermediate service and unfinished terminal capacity cannot coexist at an optimum. Before terminal capacity is exhausted, capped equalization allocates the group’s obligation. Afterward, a separable capped-cost problem allocates the remaining service. Its dependence on the menu is confined to the last eligible command.

This localization changes the joint optimization problem. We add one exact correction at the path edge that leaves an eligibility class, or at the final selected command. A conditional path algorithm then optimizes both the menu and all individual targets for supplied class means. With common catalog eligibility, the accepted root obligation already fixes the sole class mean. One polynomial-time call solves the original finite-catalog problem globally, at interior as well as saturated obligations, for any number of histories, variable menu size, heterogeneous caps and quadratic costs, and arbitrary nonnegative opening charges. No averaging error, cap guard, command augmentation, or target grid is required in this regime.

For several eligibility classes, we derive an exact price oracle on boxes of class means. The oracle combines the terminal-first correction with an increasing and a decreasing path through the common priced command scores. The second side is needed because mandatory lower targets can make commands after the price-optimal peak indispensable. Every returned upper bound covers all allowed menus in its box. Subdivision gives an original-space interval at every interruption and reaches any positive additive tolerance when completed. Its worst-case accuracy exponent depends on the number of distinct eligible prefixes, not on the number of histories or the number of cost bins. Dependence on eligibility complexity remains exponential; the common-eligibility regime is exactly polynomial and has no accuracy parameter.

The certificate is independently checkable. A feasible service allocation and a scalar price certify each pooling correction. Separate checks cover the path inequalities, eligible anchors, every branch of the target-space partition, and the original lotteries. The checker does not invoke the optimization algorithm or infer global accuracy from a solver status. Thus the large-history result is a joint optimization statement, not a conditional solution accompanied only by a Jensen bound.

Our computational study first revisits every instance in the preceding cost-heterogeneity experiment without changing its primitives. It then tests common eligibility with up to 1,024 distinct histories and stresses growing eligibility complexity at two accuracies and two node budgets. We retain all unresolved intervals, exact exhaustive references where specified, and every direct mixed-integer envelope solve. The experiment protocol precedes execution in the repository. These are declared synthetic inputs; no calibration to an operational population or implemented deployment is asserted. Historical solver times are distinguished from new measurements rather than presented as matched-machine speed comparisons.

Relation to established methods. Interval quantization, matrix searching, pricing, and branch-and-bound supply important building blocks (Wu, 1991; Nielsen and Nock, 2014; Grønlund et al., 2017; Handler and Zang, 1980; Lawler and Wood, 1966; Lemaréchal, 2001; Hübner et al., 2025). Capped separable resource allocation and its multiplier algorithms are also established (Patriksson and Strömberg, 2015; Schoot Uiterkamp et al., 2022). Our contribution is not another water-level solver or another general subdivision rule. It is the terminal-first identity and its last-eligible-command correction, which make these tools jointly optimize a shared charged alphabet without sacrificing individual constraints. The two-sided group-box identity supplies a locally exact upper oracle for the remaining eligibility means.

The earlier aggregation results address a complementary construction. Value-function aggregation and error decomposition have broad precedents (Li and Bertsekas, 2025; Fattahi, 2025), and arithmetic–harmonic mean inequalities are classical (Liao and Wu, 2015). Our retained guarded and harmonic relaxations preserve the installed menu and allow an exact original-feasible lift. Exact eligibility pooling goes further by retaining the full within-class service frontier instead of replacing it by a representative curvature. It removes the resulting dispersion allowance altogether, at the cost of evaluating the original histories within each oracle call.

Canonical contractual allocation, continuous quadratic design, Monge acceleration, budgeted compression, target-net approximation, and charge-aware recovery remain part of the mathematical contribution under their original assumptions. Table 1 distinguishes these regimes from the exact eligibility results. The main text develops the structural joint algorithm; regular appendices and the electronic companion retain the earlier proofs. A separate computational record preserves all earlier study narratives and supplies the complete per-instance measurements and reproducibility instructions.

2 Model and Canonical Allocation

A public branch j is observed with probability $\pi_j > 0$, with $\sum_j \pi_j = 1$. It has a continuation cap $b_j \in (0, 1)$ and a convex nondecreasing intermediate-service cost h_j , with $h_j(0) = 0$. The common terminal reward f is strictly increasing and concave on $[0, 1]$. Caps may repeat; write $\bar{b} = \sum_j \pi_j b_j$. An encoder selects a terminal command from one installed alphabet, and only its symbol reaches the common terminal gateway. The gateway cannot recover the observed branch, an inherited intermediate tier, or an uncharged shared random seed. Thus m distinct commands require $\lceil \log_2 m \rceil$ writable bits; read-only program constants and public-state storage are separate resources. The full institutional mechanism and saturated benchmarks are retained in the companion.

The provider selects a pre-draw tier y_j and a lottery C_j on the installed book, with operating payoff $\mathbb{E}f(C_j) - h_j(y_j)$. The accepted root obligation is $B \in [0, \bar{b}]$. Expected participation restricts the mean obligation, while a realization ceiling restricts every positive-probability draw. With a common risk tolerance $\delta \geq 0$,

$$t_j = y_j + \mathbb{E}C_j \leq b_j, \quad y_j + C_j \leq b_j + \delta \text{ almost surely}, \quad \sum_j \pi_j t_j = B. \quad (1)$$

A book c has at most m distinct levels in $[0, 1]$ and incurs $\sum_{z \in c} \rho(z)$, once per selected level. Let $\mathcal{V}_m^\delta(B; \rho)$

denote maximum operating payoff less this charge. A finite catalog restriction is denoted $\mathcal{V}_{m,\mathcal{A}}^\delta(B;\rho)$. The expectation cap is not replaced by the weaker realization ceiling. Taking $\delta \geq 1$ makes the realization restriction redundant at the canonical optimum below. Taking $\delta = 0$ gives the full pathwise value, even though that institution initially permits draw-specific intermediate service. Thus these are endpoints of one design problem, not three unrelated extensions.

2.1 Eligibility prefixes replace saturated targets

For a strictly ordered book $c = (c_1, \dots, c_s)$ and branch j , define

$$c^{j,\delta} = \{z \in c : z \leq b_j + \delta\}, \quad d_j = \max c^{j,\delta},$$

and let $F_{j,c}^\delta$ be the piecewise-affine interpolant of f on this prefix, including the singleton interpretation. The prefix depends on the cap and risk ceiling, not on the endogenous target.

Theorem 2.1 (Canonical allocation under a realization ceiling). *Suppose f is strictly increasing and concave and h_j is convex, nondecreasing, and zero at zero. A fixed book is feasible exactly when $c_1 \leq \min_j b_j$ and $c_1 \leq B \leq \bar{b}$. Its optimal branch response at target $c_1 \leq t \leq b_j$ is*

$$v_{j,c}^\delta(t) = F_{j,c}^\delta(\min\{t, d_j\}) - h_j((t - d_j)_+). \quad (2)$$

It is attained by $y = (t - d_j)_+$ and an adjacent lottery of mean $\min(t, d_j)$ in the eligible prefix. Consequently

$$\mathcal{V}_m^\delta(B;\rho) = \max_{\substack{|c| \leq m, c_1 \leq t_j \leq b_j \\ \sum_j \pi_j t_j = B}} \left\{ \sum_j \pi_j v_{j,c}^\delta(t_j) - \sum_{z \in c} \rho(z) \right\}. \quad (3)$$

The fixed-book inner problem is concave. With zero charges, $\mathcal{V}_m^0(B;0) = V_m^P(B)$ and $\mathcal{V}_m^\delta(B;0) = V_m^E(B)$ for $\delta \geq 1$, at every feasible B .

Proof. Every used terminal level satisfies $C \leq b_j + \delta$ because $y \geq 0$. Writing $\mu = \mathbb{E}C$, the payoff is at most $F_{j,c}^\delta(\mu) - h_j(t - \mu)$. Both terms are nondecreasing as μ increases on $[c_1, \min(t, d_j)]$, so this upper bound is maximized at its right endpoint. If $t \leq d_j$, the adjacent lottery has $y = 0$ and all its support lies below $b_j + \delta$. If $t > d_j$, it is the singleton d_j and $y + d_j = t \leq b_j$. Thus the upper bound is attained without relaxing either constraint. The response is concave: its nonnegative interpolation slopes decrease, and its tail has the nonpositive slopes of $-h_j(t - d_j)$. The feasibility conditions follow by using the lowest level and allocating intermediate service; weighted targets fill the entire interval $[c_1, \bar{b}]$.

For pathwise participation with possibly draw-specific Y , all used C satisfy $C \leq b_j$. At fixed $t = \mathbb{E}(Y + C)$, concavity of f on the discrete support and Jensen's inequality for h_j give the same upper bound with $\delta = 0$ and $\mathbb{E}Y = t - \mu$. The canonical policy attains it and is pathwise feasible. Fresh decoder randomization can first be replaced by its conditional mean for each symbol: this increases terminal reward and preserves all the ceilings under the stipulated absence of an uncharged shared seed. Under expected participation the

same mean argument has no eligibility truncation; $\delta \geq 1$ includes every level in $[0, 1]$. These observations establish the two endpoint identities and both directions of (3). $\square$

For quadratic primitives (56), chord $[u, v]$ has slope $r - q(u + v)/2$. On branch j its target capacity is

$$\left[\min\{b_j, v\} - u \right]_+ \mathbf{1}\{v \leq b_j + \delta\}.$$

Starting at $t_j = c_1$, fill these positive slopes in their common decreasing order, followed by zero-cost intermediate capacity and the quadratic tails. A water-level sweep over events $\gamma_j(b_j - d_j)_+$ fills the remaining targets. It uses $O(k s + k \log(k + 1))$ rational arithmetic operations, including the certificate

$$\sum_j \pi_j v_{j,c}^\delta(t_j) = \lambda B + \sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}^\delta(t) - \lambda t\}. \quad (4)$$

The right-hand branch maxima are independently checked at eligible knots, the cap, and the clipped tail stationary point. This multiplier certifies the allocation for that book; it is not by itself a global codebook certificate. The outer problem is solved next.

All canonical arguments continue to hold for branch-specific ceilings $\tau_j \geq b_j$ in place of $b_j + \delta$: the eligibility prefix is then $c \cap [0, \tau_j]$. We write $\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho)$ for this generalized catalog value. The new budgeted algorithms and numerical formulation use these heterogeneous ceilings. The retained continuous and institutional results use their stated common- δ specialization.

3 Exact Budgeted Compression at Fixed Targets

We first solve alphabet selection without changing the promised branch targets. This is both an independently meaningful design problem and an oracle for joint design. Let $\mathcal{A} = (a_1 < \dots < a_N)$, with charges $\rho_i \geq 0$, and fix $t_j \leq b_j$, $\sum_j \pi_j t_j = B$, and $a_1 \leq \min_j t_j$. Write $\tau_j \geq b_j$ for the realization ceiling; the renewal specialization is $\tau_j = b_j + \delta$. In this section $v_{j,c}$ denotes (2) with this branch-specific ceiling, and $R(c) = \sum_{a_i \in c} \rho_i$.

For two candidate levels $u < v$, define the target-crossing edge and stopping reward

$$A_t(u, v) = \sum_{j: u \leq t_j < v} \pi_j \begin{cases} f(u) + \frac{t_j - u}{v - u} \{f(v) - f(u)\}, & v \leq \tau_j, \\ f(u) - h_j(t_j - u), & v > \tau_j, \end{cases} \quad (5)$$

$$Z_t(u) = \sum_{j: t_j \geq u} \pi_j \{f(u) - h_j(t_j - u)\}. \quad (6)$$

The eligibility test is on v , not on its mean with u . If it fails, a branch uses the lower level and a fixed tier $t_j - u$; averaging an ineligible realization into a feasible mean would not implement the contract. Unlike the priced recurrence, (5) has no price-admissibility restriction on an arc.

Theorem 3.1 (Conditional memory–charge frontier). *For any feasible target vector t , define*

$$F_\ell(t) = \max_{\substack{c \subseteq \mathcal{A}, |c|=\ell \\ c_1 \leq \min_j t_j}} \left\{ \sum_j \pi_j v_{j,c}(t_j) - R(c) \right\}, \quad P_s(t) = \max_{1 \leq \ell \leq s} F_\ell(t).$$

Infeasible exact cardinalities have value $-\infty$. Set $H_1(i) = Z_t(a_i)$ and

$$H_\ell(i) = \max_{v > i} \{A_t(a_i, a_v) - \rho_v + H_{\ell-1}(v)\}. \quad (7)$$

Then $F_\ell(t) = \max_{a_i \leq \min_j t_j} \{H_\ell(i) - \rho_i\}$. All frontiers through budget $M \leq N$ and their feasible policies are computable in $O((k + M)N^2)$ arithmetic operations and $O(N^2 + MN)$ storage, given constant-time evaluations of f and h_j . Rational quadratic data give polynomial bit complexity. Every policy uses one fixed book, preserves the supplied targets exactly, commits intermediate service before its lottery, and respects every realization ceiling.

Proof. Take a feasible ordered book. Every target belongs to exactly one interval $[u, v)$ between consecutive levels, or to the final interval $[u, \infty)$. In the former case $u \leq t_j \leq b_j \leq \tau_j$, so the lower endpoint is eligible. If v is eligible, the canonical response is its chord at t_j . If v is ineligible, every later level is ineligible, and the highest eligible book level is u ; the response is the second line of (5). A final-interval branch has the response in (6). These intervals partition the branches, including targets equal to a selected level. The book’s operating value is therefore the sum of its edge and stopping rewards.

The recurrence enumerates ordered paths with exactly ℓ vertices. It pays every charge once, with the first charge outside the recurrence. Conversely, each such path starting no higher than the smallest target defines a feasible book, and the preceding partition computes its exact value. Maximization proves both directions of the claimed equality. Taking the best exact cardinality gives an at-most budget and does not assume that an unused additional symbol should be purchased. There are $O(N^2)$ edges, each computed by scanning the branches, and $O(MN^2)$ transitions. Each rational path value is a sum of polynomially many input-derived rational terms, so ordinary rational arithmetic and comparison have polynomial bit complexity. $\square$

Corollary 3.2 (Global catalog design at saturation). *If $B = \bar{b}$, then $P_s(b) = \mathcal{V}_{s,\mathcal{A}}^\tau(\bar{b}; \rho)$ for every s . Hence joint catalog design at a saturated promise, including arbitrary nonnegative opening charges and branch-specific realization ceilings, has the complexity in Theorem 3.1, with both k and s variable.*

Proof. Positive weights and $t_j \leq b_j$ imply $\sum_j \pi_j t_j = \bar{b}$ only when every $t_j = b_j$. There is no remaining target optimization. Apply the theorem. $\square$

This corollary is a same-budget global result, not a priced relaxation. At an interior promise, the theorem optimizes the entire alphabet conditional on t ; it does not certify that the selected target vector is jointly optimal. Section A resolves that target choice at fixed branch count. The faster saturated Monge construction in Appendix E concerns its stronger structural assumptions and is retained rather than replaced.

3.1 Compression, opening costs, and symbol prices

Take any candidate pool U , including a union of supported books. Applying the theorem with $\mathcal{A} = U$ gives the optimal sub-book at every allowed cardinality, not just a sequence of greedy deletions. Applying it on the whole catalog can only improve the conditional value. No nestedness of optimal books is required. In particular, comparing budgets $m, m + 1, \dots, |U|$ gives the complete conditional memory–value frontier and identifies the value purchased by each additional symbol.

If an additional uniform certification fee $\kappa \geq 0$ is imposed, the exact conditional value is

$$\max_{1 \leq \ell \leq M} \{F_\ell(t) - \kappa \ell\}. \quad (8)$$

The upper envelope has finitely many breakpoints, obtained from the computed exact-cardinality values. Chosen cardinalities may be tied, but a largest optimal cardinality is nonincreasing in κ . Indeed, optimality at $\kappa_1 < \kappa_2$ and cardinalities ℓ_1, ℓ_2 gives $(\kappa_2 - \kappa_1)(\ell_1 - \ell_2) \geq 0$. This statement concerns cardinality, not nesting of the selected levels. General nonuniform charges are already included in the recurrence and do not require the uniform-fee interpretation.

The structural advantage over a raw union is that charges enter the optimization itself. A costly level may be discarded even when capacity remains. Nevertheless, an optimal conditional compression can be inferior to a policy with different branch targets. Our returned operational policy therefore compares valid candidates at the *same* budget and keeps the best, including a supplied incumbent. This safeguard never converts a heuristic target portfolio into a global optimality theorem.

4 Exact Pooling by Eligibility

Two histories have the same *eligibility class* when they admit the same catalog prefix. Let $G_1, \dots, G_E$ be these classes, and write

$$W_g = \sum_{j \in G_g} \pi_j, \quad w_{j|g} = \pi_j / W_g, \quad \bar{b}_g = \sum_{j \in G_g} w_{j|g} b_j.$$

This partition does not require identical caps or costs. In particular, an arbitrarily large collection with common catalog eligibility has $E = 1$, even if every cap and every curvature differs. We retain each original branch when evaluating a class; we do not replace it by an artificial branch with an average cost. The group mean s_g satisfies $\sum_g W_g s_g = B$.

For $0 \leq s \leq \bar{b}_g$, define the capped equalization

$$\theta_j^g(s) = \min\{b_j, z_g(s)\}, \quad \sum_{j \in G_g} w_{j|g} \min\{b_j, z_g(s)\} = s. \quad (9)$$

At $s = \bar{b}_g$, use $\theta_j^g(s) = b_j$. Although the water level is then not unique, the allocation is unique. Each coordinate is nondecreasing in s . For a book c with a feasible anchor, let d_g be its last level eligible for this

class and let $F_{g,c}$ be its common reward interpolant. Define

$$D_g(d) = \sum_{j \in G_g} w_{j|g} \min\{b_j, d\}, \quad (10)$$

$$C_g(q; d) = \min \left\{ \sum_{j \in G_g} w_{j|g} h_j(y_j) : 0 \leq y_j \leq (b_j - d)_+, \sum_{j \in G_g} w_{j|g} y_j = q \right\}. \quad (11)$$

The second problem is ordinary separable resource allocation, not a new water-filling algorithm. Breakpoint and multiplier methods for such problems are established; see Patriksson and Strömberg (2015) and Schoot Uiterkamp et al. (2022). What matters here is that its only book-dependent argument is the last eligible level.

Theorem 4.1 (Terminal-first pooling). *For continuous convex nondecreasing costs that vanish at zero, the maximum normalized payoff of class g at mean $s \in [c_1, \bar{b}_g]$ equals*

$$V_{g,c}(s) = \begin{cases} \sum_{j \in G_g} w_{j|g} F_{g,c}(\theta_j^g(s)), & s \leq D_g(d_g), \\ \sum_{j \in G_g} w_{j|g} F_{g,c}(\min\{b_j, d_g\}) - C_g(s - D_g(d_g); d_g), & s \geq D_g(d_g). \end{cases} \quad (12)$$

Consequently,

$$W_g V_{g,c}(s) = \sum_{j \in G_g} \pi_j v_{j,c}^\tau(\theta_j^g(s)) + \Delta_g(d_g; s), \quad (13)$$

$$\Delta_g(d; s) = \sum_{j \in G_g} \pi_j h_j((\theta_j^g(s) - d)_+) - W_g C_g((s - D_g(d))_+; d) \geq 0. \quad (14)$$

Below $D_g(d_g)$ an optimal allocation is $t_j = \theta_j^g(s)$; above it an optimal allocation is $t_j = \min\{b_j, d_g\} + y_j$, where y solves (11). These policies satisfy the individual caps and realization ceilings, not merely their group averages.

Proof. Use the canonical response in Theorem 2.1. Suppose a feasible allocation gives positive intermediate service to one member while another has unused terminal capacity below $\min\{b_j, d_g\}$. Transfer a small positive amount of weighted target mass from the first member to the second. Intermediate cost cannot increase, and the strictly increasing common interpolant raises terminal reward. Thus no optimum mixes positive service with unfinished terminal capacity.

If $s \leq D_g(d_g)$, an optimum therefore uses no service. Maximizing the same concave interpolant over capped targets gives (9). To verify this without assuming differentiability, take a supporting slope at the common unsaturated level. On an unsaturated member its supporting-line inequality bounds any change in reward. On a capped member, decreasing its target loses at least that slope per unit, by concavity. The weighted linear terms sum to zero. At the endpoint where all terminal capacities are full, the assertion follows directly by monotonicity. This proves the first case of (12).

If $s \geq D_g(d_g)$, every terminal capacity is full and only service remains to be allocated. Its capacities and total are precisely those in (11), proving the second case. For these means, $\theta_j^g(s) \geq \min\{b_j, d_g\}$, so

the equalized allocation has the same terminal rewards as the optimal allocation. Their difference is exactly the reduction in service cost in (14). Below $D_g(d_g)$ both cost terms are zero. Feasibility follows from the original cap constraints and the canonical adjacent lottery or singleton, including its supportwise ceiling. $\square$

4.1 Joint book selection at fixed group means

Let K_g be the number of catalog levels eligible for class g . For supplied group means, form the individual targets $t_j = \theta_j^g(s_g)$ and the target-crossing edges and tails in Section 3. Add

$$\sum_{g:i \leq K_g < v} \Delta_g(a_i; s_g) \quad (15)$$

to the edge from catalog index i to v , and add $\sum_{g:i \leq K_g} \Delta_g(a_i; s_g)$ to a path ending at i . Indices in this formula are one-based. The correction is paid at the unique edge that leaves a class's eligible prefix, or at the terminal node if no such edge occurs. All original level-opening charges remain in the path objective.

Theorem 4.2 (Exact eligibility-mean frontier). *The corrected path recurrence returns the globally best book and individual targets for supplied feasible group means, at every budget up to m . For rational quadratic primitives $f(x) = rx - qx^2/2$ and $h_j(y) = \gamma_j y^2/2$, including zero service curvatures, one call takes*

$$O((k + m)N^2 + kN \log(k + 1)) \quad (16)$$

rational arithmetic operations, with polynomial bit complexity in the instance and supplied means. When $E = 1$, set $s_1 = B$: a single call solves the original joint finite-catalog problem globally, for variable k, N, m , arbitrary nonnegative opening charges, and heterogeneous caps and curvatures. It does not enumerate books or expand the command budget.

Proof. For any book, the uncorrected path sums its responses at the equalized targets. Every class has exactly one last eligible selected symbol. Formula (15) adds its correction exactly once, so Theorem 4.1 makes the path value equal to that book's optimal value at the group means. Conversely every path defines a legal book and the theorem constructs its optimal individual allocation. Maximizing paths therefore proves both inequalities needed for exactness.

For each possible last level, solve (11) by first filling zero-cost capacity and then sorting the positive breakpoints $\gamma_j(b_j - d)_+$. The remaining water level solves a linear equation between consecutive breakpoints. This requires $O(|G_g| \log(|G_g| + 1))$ arithmetic operations per class and level. Compute equalized targets once, construct the original $O(kN^2)$ crossing terms, and add all class corrections in $O(EN^2)$ operations. Since $E \leq k$, the path recurrence and these costs give (16). All sums, comparisons, divisions, and reconstructed probabilities are rational with polynomial encoding length. For $E = 1$ the root promise already fixes the only group mean. Thus there is no outer target search, including at interior promises. $\square$

This result differs from aggregation of identical response types in Proposition 5.1. It retains arbitrary within-class heterogeneity and evaluates its exact service frontier. It also differs from harmonic coarsening:

no cap guard or dispersion allowance is needed. The earlier harmonic and minimum-curvature results remain useful simpler upper models under their stated assumptions and are retained in the companion.

5 Response Types and the Complexity Map

Write $P_m(B) = \mathcal{V}_{m, \mathcal{A}}^\tau(B; \rho)$ and $R(c) = \sum_{a_i \in c} \rho_i$. All results in this section concern a fixed finite catalog $\mathcal{A} = (a_1 < \dots < a_N)$. A response type consists of a cap, a shortfall cost function, and an eligible catalog prefix. Let d denote the number of such types; this is distinct from the quadratic reward-curvature parameter q in the continuous-design appendix.

Proposition 5.1 (Exact response-type reduction). *Partition branches into groups with the same b_j , the same h_j on $[0, b_j]$, and the same set $\mathcal{A} \cap [0, \tau_j]$. Replace each group by one branch whose weight is the sum of its weights and whose ceiling is the minimum of its ceilings. The reduced and original joint catalog problems have the same optimum at every feasible B and every budget m . A reduced solution lifts by assigning its target and canonical lottery to every branch in its group.*

Proof. Every book is a subset of $\mathcal{A}$. Group members therefore have exactly the same eligible book and the same concave response v_c . For a group of total weight w , replace its targets by their weighted average $\bar{t}$. This preserves its contribution to the root equality, keeps the average between the common anchor and cap, and gives $wv_c(\bar{t}) \geq \sum_{j \text{ in group}} \pi_j v_c(t_j)$. Neither the book nor its charge changes. Conversely, the common reduced target and its canonical realization satisfy every original ceiling because the eligible catalog prefix is unchanged. The two feasible mappings prove equality of optimal values. $\square$

The common-ceiling homogeneous aggregation argument already appears in the retained continuous analysis. The present statement uses catalog-prefix equivalence and permits distinct numerical ceilings. Equal caps alone do not suffice when either the cost or eligible prefix differs. Forming rational quadratic signatures by binary searching the ordered catalog takes $O(k \log N)$ prefix comparisons, plus grouping and output work. With hashing, grouping uses expected linear dictionary operations; comparison-based sorting instead adds $O(k \log k)$ comparisons. General cost-function equality requires a supplied exact representation or a verified type partition.

Proposition 5.2 (Positive complexity regimes). *For rational quadratic primitives: (i) one response type or $B = \bar{b}$ is solved by one invocation of Theorem 3.1; (ii) for every fixed m , exact joint catalog design is polynomial in N , k , and input bit length by enumerating at most $\sum_{s=1}^m \binom{N}{s}$ books and applying the fixed-book allocator; and (iii) the adaptive method below has polynomial work per price per node and a finite additive scheme in dimension $d - 1$.*

Proof. With one type the root equality fixes its target. At saturation positive weights force every target to its cap. For a specified book the concave allocation and its supporting-price certificate are polynomial for rational quadratic data, so enumerating the stated finite collection proves (ii). Theorems 6.1 and 7.1 prove (iii). $\square$

Enumeration is an XP bound in m , not a fixed-parameter bound of the form $F(m)N^c$ with a uniform exponent. A hardness assertion with a fixed small m would contradict (ii) unless P equals NP. We establish neither general NP-hardness nor a polynomial-time algorithm when both m and the number of distinct types vary. The additive scheme below remains exponential in d in its worst case. This map distinguishes proved upper bounds from unresolved lower bounds rather than treating the earlier grid's dimension as an intrinsic impossibility.

6 An Exact Two-Sided Price Oracle

After any exact type reduction, retain the index j for a branch and its weight π_j . A target box $X = [\ell, u]$ satisfies $0 \leq \ell_j \leq u_j \leq b_j$ and intersects $\sum_j \pi_j t_j = B$. Only anchors in

$$I_X = \left\{ i : a_i \leq \min_j u_j, \quad \sum_j \pi_j \max\{\ell_j, a_i\} \leq B \right\} \quad (17)$$

can meet the equality inside this box. Because $B \leq \sum_j \pi_j u_j$, these conditions are also sufficient for a fixed book's box feasibility. They depend on its lowest level, not on its remaining levels.

Let $g(x) = f(x) - \lambda x$ at a fixed real price λ . For a book whose first index lies in I_X , relax only the aggregate equality and define

$$S_X(\lambda) = \max_{\substack{\emptyset \neq c \subseteq \mathcal{A}, |c| \leq m \\ c_1 = a_i, i \in I_X}} \left\{ \sum_j \pi_j \max_{\max\{\ell_j, c_1\} \leq t \leq u_j} [v_{j,c}(t) - \lambda t] - R(c) \right\}, \quad (18)$$

$$U_X(\lambda) = \lambda B + S_X(\lambda). \quad (19)$$

Every $U_X(\lambda)$ bounds the best original-budget joint design in X . No zero-duality-gap assumption is made. For a general cost define the clipped scalar support

$$\Phi_j(s, t; \lambda) = \max_{s \leq y \leq t} \{-h_j(y) - \lambda y\}.$$

For quadratic costs it is obtained by clipping $-\lambda/\gamma_j$ to $[s, t]$ when $\gamma_j > 0$; for zero curvature choose the endpoint with the better linear value. Write $\psi_j(z) = \Phi_j(\max\{0, \ell_j - z\}, u_j - z; \lambda)$ whenever $z \leq u_j$.

6.1 The increasing and decreasing sides

For $x < z$ define an upper-crossing edge and its terminal value by

$$A^+(x, z) = \sum_{j: x \leq u_j < z} \pi_j \begin{cases} g(x) + \frac{u_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) + \psi_j(x), & z > \tau_j, \end{cases} \quad (20)$$

$$T^+(x) = \sum_{j: u_j \geq x} \pi_j [g(x) + \psi_j(x)]. \quad (21)$$

When $\lambda > 0$ also define a lower-crossing edge, a lower terminal value, and a peak value:

$$A^-(x, z) = \sum_{j: x \leq \ell_j < z} \pi_j \begin{cases} g(x) + \frac{\ell_j - x}{z - x} [g(z) - g(x)], & z \leq \tau_j, \\ g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x), & z > \tau_j, \end{cases} \quad (22)$$

$$T^-(x) = \sum_{j: \ell_j \geq x} \pi_j [g(x) - h_j(\ell_j - x) - \lambda(\ell_j - x)], \quad (23)$$

$$H(x) = \sum_{j: \ell_j < x \leq u_j} \pi_j g(x). \quad (24)$$

Only increasing-score arcs use A^+ , and only decreasing-score arcs use A^- . Equal-score arcs may be assigned to either side. Set missing forward states to $-\infty$, and compute

$$\alpha_1(i) = \begin{cases} -\rho_i, & i \in I_X, \\ -\infty, & \text{otherwise,} \end{cases} \quad (25)$$

$$\alpha_r(v) = -\rho_v + \max_{\substack{i < v \\ g(a_i) \leq g(a_v)}} [\alpha_{r-1}(i) + A^+(a_i, a_v)], \quad 2 \leq r \leq m.$$

This table pays the charge for its last level. For positive prices a backward table excludes the charge for its first level:

$$\beta_1(i) = T^-(a_i), \quad (26)$$

$$\beta_s(i) = \max \left\{ T^-(a_i), \max_{\substack{v > i \\ g(a_v) \leq g(a_i)}} [A^-(a_i, a_v) - \rho_v + \beta_{s-1}(v)] \right\}, \quad 2 \leq s \leq m.$$

Theorem 6.1 (Two-sided box-price decomposition). *If I_X is nonempty, the exact support in (18) is*

$$S_X(\lambda) = \begin{cases} \max_{1 \leq r \leq m, i} \{\alpha_r(i) + T^+(a_i)\}, & \lambda \leq 0, \\ \max_{1 \leq r \leq m, i} \{\alpha_r(i) + H(a_i) + \beta_{m-r+1}(i)\}, & \lambda > 0. \end{cases} \quad (27)$$

The maximizing book and scalar target supports are recoverable. Computing one price bound takes $O((d+m)N^2)$ arithmetic operations and $O(N^2 + mN)$ storage, where d is the number of branches after type reduction. General costs additionally require $O(dN^2)$ clipped scalar-oracle calls. Rational quadratic data and a rational price give an exact rational bound in polynomial bit complexity for each node.

Proof. Concavity of g implies that its values on any ordered book increase to a peak and then decrease; a tie can be placed on either side. For $\lambda \leq 0$, strict increase of f makes all scores strictly increasing, so the peak is the final level. Consider first a positive price and a chosen peak p .

If $u_j < p$, exactly one increasing edge $[x, z)$ crosses u_j . If z is eligible, the interpolant of g is nondecreasing through this edge and is maximized on the permitted target interval at u_j . No later segment below u_j exists. If z is ineligible, every later book level is ineligible, so x is the last usable level and clipped

shortfall maximization gives $g(x) + \psi_j(x)$. This proves the upper-crossing formula. Its clipping is essential when $\ell_j > x$.

If $\ell_j < p \leq u_j$, the peak is eligible because $p \leq u_j \leq b_j \leq \tau_j$. It attains $g(p)$ at zero shortfall. All eligible chord values are at most $g(p)$, and positive-price shortfall cannot improve them. This proves the middle term.

If $\ell_j \geq p$, the interpolant on the decreasing side is nonincreasing. Each shortfall tail is also nonincreasing because h_j is nondecreasing and $\lambda > 0$. The smallest permissible target, ℓ_j , therefore maximizes the branch's priced payoff. Its crossing edge gives (22); when the next level is ineligible it uses the singleton plus the required shortfall. Targets beyond the last level give (23). Equality at the peak belongs to this third group, preventing double counting.

These three groups partition the branches. Upper crossings before the peak, the peak, and lower crossings after the peak thus decompose the exact priced value of every book, not merely a dominant subset of books. Forward states enumerate all increasing prefixes starting at a feasible anchor. Backward states enumerate all decreasing suffixes, with an optional stop. Gluing a prefix of r levels to a suffix of at most $m - r + 1$ levels counts the shared peak once, counts its charge once, and uses at most m levels. Every feasible ordered book has such a representation, and every represented path is a feasible book for the box. This proves the positive-price formula.

For nonpositive prices, the best point on an eligible increasing chord up to the cap is the cap. Where the next level is ineligible, the last singleton plus clipped shortfall gives the scalar optimum; the same argument gives the final tail. Hence (20) and (21) partition all branches, proving the first case.

The edge construction scans at most d branches for each ordered pair. Each table has mN states and at most N transitions per state. Backpointers reconstruct the book, and scalar maximization reconstructs its supports. For quadratic rational inputs each edge uses a polynomial number of rational operations; each path sums polynomially many terms. Their numerator and denominator lengths remain polynomial in the input, price, and box-endpoint bit lengths. This is a per-node bit-complexity statement, not a bound independent of search depth. $\square$

Why a second side is necessary. Take equal weights, targets and caps $(1/5, 1/2, 4/5)$, unit shortfall curvature, unit ceilings, catalog $\{1/5, 1/2, 4/5\}$, zero charges, budget three, and reward $f(x) = 2x - x^2/2$. At price $3/2$, the complete book has scores $(2/25, 1/8, 2/25)$. On the singleton target box its exact joint value is $169/200$. Restricting to nondecreasing-score books gives only $83/100$, a shortfall of $3/200$. Thus the unrestricted deletion argument in Theorem B.1 cannot be reused on a target box. The new recurrence repairs that structural obstruction rather than placing the old one-sided oracle inside a generic partitioning procedure.

7 Adaptive Joint Design and Verifiable Global Intervals

Let $\Theta = \{t \in [a_1, b] : \sum_j \pi_j t_j = B\}$. A box can be reduced without losing any point of its intersection with Θ . With $s_\ell = \sum_j \pi_j \ell_j$ and $s_u = \sum_j \pi_j u_j$, its exact coordinate projections are

$$\ell'_j = \max \left\{ \ell_j, \frac{B - s_u + \pi_j u_j}{\pi_j} \right\}, \quad u'_j = \min \left\{ u_j, \frac{B - s_\ell + \pi_j \ell_j}{\pi_j} \right\}. \quad (28)$$

The box is discarded only if $B \notin [s_\ell, s_u]$ or an interval is empty. Formula (28) follows by allocating all other coordinates to their upper or lower endpoints; continuous interval filling proves attainability of each projected endpoint.

At each retained box compute $U_X = \min_{\lambda \in \Lambda} U_X(\lambda)$ for any finite price set containing zero. Every recovered book has an anchor in I_X and hence has a feasible allocation satisfying the original equality. Optimize that book's targets exactly, either within X or over the entire original target region, and update the feasible incumbent L . The implementation uses global fixed-book allocation and caches repeated books. This changes neither the original budget nor the charge schedule. Root price bisection is a choice of bound-strengthening prices, not an assertion of zero duality gap or a prerequisite for validity.

Retain the leaves of a binary cover of Θ , select a leaf with largest upper bound, and bisect one independent target coordinate at its midpoint. Apply (28) to both children. Their intersections with Θ cover their parent's intersection; the common boundary may overlap. A dependent coordinate j_0 is recovered through the equality. Use as splitting coordinate the longest weighted side $\pi_j(u_j - \ell_j)$ among $j \neq j_0$. The code chooses a largest-weight coordinate as j_0 ; this choice is not needed for validity.

Theorem 7.1 (Adaptive original-budget certificate and finite accuracy). *Assume f and all h_j are Lipschitz on their relevant compact intervals and put $L_v = \max\{L_f, \max_j L_{h,j}\}$. Use the exact oracle in Theorem 6.1, include price zero at every node, and evaluate the returned zero-price book by a feasible exact fixed-book allocation. At every interruption,*

$$L \leq P_m(B) \leq U = \max\{L, \max_{X \text{ a leaf}} U_X\}. \quad (29)$$

For every $\varepsilon > 0$, best-upper-bound selection and longest-weighted-side bisection terminate with $U - L \leq \varepsilon$. If $d = 1$ the root is a singleton and is exact. For fixed $d > 1$, a conservative full-tree bound is $2^{H+1} - 1$ nodes, where

$$H = \sum_{j \neq j_0} \max \left\{ 0, \left\lceil \log_2 \frac{2L_v(d-1)w_j^0}{\varepsilon} \right\rceil \right\}, \quad w_j^0 = \pi_j(u_j^0 - \ell_j^0). \quad (30)$$

A zero width contributes zero. With $L_v = 0$, one root evaluation suffices. The bound is polynomial in numerical $1/\varepsilon$ for fixed d , and exponential in d in general; adaptive pruning can finish before this worst-case cover is constructed.

Proof. A feasible policy supplies the lower bound. Weak duality and the complete leaf cover supply the upper bound, irrespective of the search order or the quality of the incumbent. Thus pruning is valid whenever a leaf bound is at most $L + \varepsilon$, and retaining such leaves in a certificate is harmless.

For local tightness, let c^0 maximize $S_X(0)$ and let $\hat{t}$ be its independently maximizing box targets. Condition (17) gives some equality-feasible t^0 for that same book in X . The canonical response is continuous and piecewise concave with slopes bounded in absolute value by L_v : its chord slopes are bounded by L_f and its shortfall slopes by $L_{h,j}$. Charges cancel for the same book. Therefore

$$U_X \leq U_X(0) \leq \sum_j \pi_j v_{j,c^0}(t_j^0) - R(c^0) + L_v \sum_j \pi_j (u_j - \ell_j) \leq L + L_v \sum_j \pi_j (u_j - \ell_j). \quad (31)$$

An exact global fixed-book allocation can only improve the displayed feasible value. The inequality does not require a uniform bound on a recovery-compression penalty and does not mix two books.

After coordinate projection, the dependent weighted width is no larger than the sum of the other weighted widths. To see this, choose equality-feasible points attaining its lower and upper endpoints and subtract their equalities. Thus the last sum in (31) is at most twice the sum of the independent weighted widths. No node needs further splitting once each independent width is at most $\varepsilon/[2L_v(d-1)]$. Before this occurs, a longest-width split halves a width above that threshold; projection cannot increase any width. Along a path, coordinate j can be split at most its summand in (30) before falling below the threshold. Hence every path needing refinement has at most H splits. The complete binary tree bound follows. At a singleton box the same calculation gives equality of the box upper bound and the fixed-target optimum. This also proves the one-type case. $\square$

The result replaces completion of a full target net by certificate-directed refinement. It does not promise that refinement is inexpensive for every input. It supplies a structural polynomial node oracle and a meaningful interval when it is not. An absolute tolerance is measured in payoff units. Our quadratic experiments normalize target levels to $[0, 1]$ and use $r = 2$, $q = 1$; they report both the absolute gap and its ratio to $r + \max_j \gamma_j/2 + \sum_{i=1}^m \rho_{(i)}$, where the charges are sorted decreasingly. This positive declared scale remains meaningful when net value is close to zero. Rescaling all rewards and charges requires rescaling the requested tolerance.

7.1 Independent checking and physical implementation

A certificate contains the original inputs, the exact type partition, an original-space lottery, every binary split, the authoritative leaves, and one upper-bounding price table per node. It need not contain every price tried. The checker verifies anchor bounds, every reachable forward Bellman inequality, every backward stopping and transition inequality, and the final maximum in (27). Upper tables need not equal the optimizer's tables; satisfying these inequalities is enough to dominate every represented path. It separately reconstructs both projected children at every split and verifies that no leaf is missing. Finally, it checks probabilities, selected charges, pre-draw service, the exact root promise, and supportwise realization limits without using the canonical response formula. The certificate therefore certifies all target regions, not just the sampled incumbents. Its storage is $O(TmN + Td)$ rational entries for T recorded nodes, apart from the policy and input data; the checker recomputes edge values and has polynomial work per recorded node.

The terminal alphabet is only one physical resource. A type-reduced implementation additionally stores

a history-to-type encoder, a target and lottery table, command constants, and probability representations. Arbitrary rational probabilities may require variable-length random-bit generation or a separate finite-precision error analysis; we do not count them as free writable memory. Offline certificates have the size just stated and are not assumed to fit in the terminal gateway. The total deployment cost is consequently not identified with $\lceil \log_2 m \rceil$ bits. This separation is essential to interpreting the repeated-type computation as joint design of a restricted interface, rather than a claim about full-system memory or calibrated field performance.

8 Certified Search in Eligibility Means

For several eligibility classes, the joint search can use their means instead of individual targets. Let $X = [\ell, u]$ be a group-mean box intersecting $\sum_g W_g s_g = B$, with $0 \leq \ell_g \leq u_g \leq \bar{b}_g$. A feasible book anchor a must satisfy

$$a \leq \min_j b_j, \quad a \leq \min_g u_g, \quad \sum_g W_g \max\{\ell_g, a\} \leq B. \quad (32)$$

These conditions are sufficient as well: all individual caps are at least a , and the group-mean intervals fill every total between their lower and upper sums. Define the artificial individual box

$$l_j = \theta_j^g(\ell_g), \quad v_j = \theta_j^g(u_g), \quad j \in G_g. \quad (33)$$

Its feasible-anchor test is equivalent to (32). The artificial box is only an algebraic input to the existing two-sided oracle; we correct its priced value before treating it as a group-box bound.

8.1 The exact service correction at a price

Fix a real price λ and a possible last eligible level d . If $u_g < D_g(d)$, set $\Xi_g(d; X, \lambda) = 0$. Otherwise, put

$$q_- = (\ell_g - D_g(d))_+, \quad q_+ = u_g - D_g(d), \quad c_j = (b_j - d)_+,$$

and define

$$T_g(d; X, \lambda) = \max_{\substack{0 \leq y_j \leq c_j \\ q_- \leq \sum_j w_{j|g} y_j \leq q_+}} \sum_{j \in G_g} w_{j|g} \{-h_j(y_j) - \lambda y_j\}, \quad (34)$$

$$J_g(d; X, \lambda) = \sum_{j \in G_g} w_{j|g} \max_{(l_j - d)_+ \leq y \leq (v_j - d)_+} \{-h_j(y) - \lambda y\}, \quad (35)$$

$$\Xi_g(d; X, \lambda) = W_g [T_g(d; X, \lambda) - J_g(d; X, \lambda)]. \quad (36)$$

Every vector admitted by J_g has mean service in $[q_-, q_+]$. Thus $\Xi_g \geq 0$. Quadratic T_g is computed by clipping its unconstrained total to $[q_-, q_+]$ and solving the capped-cost problem at that total.

Theorem 8.1 (Exact group-box price decomposition). *For every book with an anchor satisfying (32),*

$$W_g \max_{\max\{\ell_g, c_1\} \leq s \leq u_g} \{V_{g,c}(s) - \lambda s\} = \sum_{j \in G_g} \pi_j \max_{\max\{l_j, c_1\} \leq t \leq v_j} \{v_{j,c}^\tau(t) - \lambda t\} + \Xi_g(d_g; X, \lambda). \quad (37)$$

Apply the two-sided recurrence of Theorem 6.1 to (33). Add $\sum_{g:i \leq K_g < v} \Xi_g(a_i; X, \lambda)$ to both its rising and falling edges, and add $\sum_{g:i \leq K_g} \Xi_g(a_i; X, \lambda)$ to both terminal formulas. Leave its peak term and charges unchanged. The resulting support $S_X^{\text{elig}}(\lambda)$ is exact over all original-budget books and group-box targets. Hence

$$U_X^{\text{elig}}(\lambda) = \lambda B + S_X^{\text{elig}}(\lambda) \quad (38)$$

is a global upper bound for the joint problem in X , with the per-call arithmetic bound (16) for rational quadratic primitives.

Proof. If $u_g < D_g(d_g)$, all useful means lie in the terminal-only range. Capped equalization in Theorem 4.1 attains the group optimum at every such mean, and its coordinates lie in the artificial box. Conversely every vector in that box has an allowed group mean and is an original feasible allocation. The two priced suprema are equal and no correction is needed.

Suppose $u_g \geq D_g(d_g)$. For $\lambda \leq 0$, terminal reward minus price is strictly increasing. Every member's terminal capacity is therefore filled before service is chosen. The true group problem has residual service interval $[q_-, q_+]$, whereas the artificial box has the separate service bounds in J_g . Their terminal reward and price constants coincide; their difference is (36).

For $\lambda > 0$ and $\ell_g \leq D_g(d_g)$, neither problem needs service at an optimum. Reducing service strictly improves the priced objective, and an allowed terminal-only allocation exists. The capped-equalization argument again gives equality of the priced suprema; both T_g and J_g are zero. Finally, if $\lambda > 0$ and $\ell_g > D_g(d_g)$, service is unavoidable. Each problem chooses its smallest allowed mean. Their terminal parts coincide and their service difference is again $T_g - J_g$. The anchor cannot alter a service interval in these cases because $c_1 \leq d_g$ and $c_1 \leq \min_j b_j$.

This proves (37) for every sign and boundary case, including $\lambda = 0$. As in Theorem 4.2, the last eligible level is assigned to exactly one crossing edge or terminal node. The two-sided recurrence represents every book through the unimodality of the common level scores, so the added correction changes its value without restricting its paths. Weak duality gives (38). Computing all service corrections costs $O(kN \log(k+1))$ operations; the inherited edge and path costs give the stated bound. $\square$

8.2 Accuracy and independent certificates

The group response $V_{g,c}$ is L -Lipschitz in its mean whenever all individual responses are L -Lipschitz. Indeed, increasing or decreasing the mean of a feasible original allocation by η can be implemented by bounded transfers in one direction with weighted absolute change $|\eta|$. Applying the individual Lipschitz bounds and reversing the comparison proves the claim. For the quadratic model one may take $L = \max\{r, \max_j \gamma_j\}$.

Theorem 8.2 (Eligibility-parameterized joint accuracy). *For $E > 1$ and $\varepsilon > 0$, exact conditional optimization on a repaired net of group means returns an original-feasible policy with loss at most ε . A sufficient number of oracle calls is*

$$N \left(1 + \frac{2L(E-1)}{\varepsilon} \right)^{E-1}. \quad (39)$$

Alternatively, projected group-box subdivision using (38), with zero included among the tested prices and exact recovery of each returned book, gives a valid rational interval at every interruption. A completed finite subdivision attains any positive tolerance. Its worst-case number of leaves is at most

$$\left(\max \left\{ 1, \frac{4L(E-1)}{\varepsilon} \right\} \right)^{E-1} \quad (40)$$

when the largest weighted free-coordinate width is bisected; the number of nodes is less than twice the number of leaves. For $E = 1$, Theorem 4.2 gives an exact single-call solution instead of either search.

Proof. For each feasible catalog anchor a , discretize $E - 1$ weighted group means $W_g s_g$ in their intervals $[W_g a, W_g \bar{b}_g]$ with step $\eta = \varepsilon / [2L(E - 1)]$. Set the final mean to the remaining promise, clip it to its bounds, and repair any residual by bounded transfers. These are operations on group means; Theorem 4.2 subsequently allocates the original branches exactly.

Consider an optimal book with anchor a and means s^* . Rounding down its free weighted coordinates removes less than $(E - 1)\eta$ total mass. The dependent coordinate and subsequent upward repair restore precisely that mass without leaving the group caps. The weighted absolute distance from s^* is therefore less than $2(E - 1)\eta$. The same optimal book remains feasible. The group Lipschitz bound limits its loss to ε , and the exact conditional oracle cannot do worse. Each free coordinate has weighted range at most one; enumeration of anchors gives (39).

For subdivision, let $w(X) = \sum_g W_g (u_g - \ell_g)$. An optimal zero-price relaxed book has group targets in X . Its anchor satisfies (32), so bounded transfers can restore the root equality within the same box and book. The loss is at most $Lw(X)$. Exact allocation of that returned book at the root promise does at least as well. Therefore a node cannot retain a certification gap greater than $Lw(X)$.

Project each box onto the root equality and choose one dependent coordinate. Its weighted width is no larger than the sum of the free weighted widths. Consequently $w(X)$ is at most twice that sum. If all free widths are at most η , the node gap is at most ε . A node needing further subdivision has a free width exceeding η ; bisecting the largest one never requires more than $\max\{0, \lceil \log_2(1/\eta) \rceil\}$ cuts along any coordinate of a root-to-leaf path. This bounds the full binary cover by (40). The covering argument and weak duality remain valid before completion. $\square$

The exponent is now the number of distinct eligible prefixes minus one, not the number of histories or accuracy-dependent cost bins. At fixed E and bounded primitive scale, this is an additive scheme polynomial in the history count, catalog size, and inverse accuracy. It is not a relative-error approximation assertion, nor a polynomial bound when eligibility complexity varies. The exact common-eligibility result is stronger: it needs no tolerance parameter at all.

For independent checking, a service allocation in (34) is accompanied by one rational multiplier ν . Its upper witness is

$$T_g \leq \nu \begin{cases} q_+, & \nu \geq 0, \\ q_-, & \nu < 0 \end{cases} + \sum_{j \in G_g} w_{j|g} \max_{0 \leq y \leq c_j} \{-h_j(y) - (\lambda + \nu)y\}. \quad (41)$$

The checker evaluates each scalar maximum at endpoints and the clipped quadratic stationary point. Equality with the supplied feasible service vector verifies the correction exactly. It then checks both path-table inequalities, every feasible anchor, the complete binary group-box cover, and the original uneliminated policy. It does not import the pooling optimizer or trust a completion flag. This closes the chain from an exact service allocation to an original-space joint-design interval.

9 Computational Study

The pre-execution protocol specifies 64 exact-rational runs and 36 direct mixed-integer envelope solves. All inputs are synthetic, on the normalized domain $[0, 1]$, with common reward $f(x) = 2x - x^2/2$. We evaluate the algorithm in Section 8, not a conditional target portfolio. Every reported interval uses an original-feasible contract and a complete checked upper cover. The computational record gives the generators, all raw cases, actual toolchain, certificate hashes, and independent checking times. Historical studies are preserved without reclassifying their conditional or heuristic outputs as global guarantees.

Paired comparison without changing the contracts. We use all 24 preceding harmonic-coarsening instances, including distinct caps and costs, zero curvature, positive opening charges, and up to 256 histories. The budget, catalog, root promise, and individual constraints are unchanged. Each new run has tolerance 10^{-3} , at most 31 nodes, and four root-price refinements. These are the same node and price budgets as the earlier paired study; timings are not claimed to be matched across execution environments.

All 24 new intervals meet the requested tolerance, compared with 14 of the 24 retained harmonic intervals. Nineteen new intervals have zero rational width. The largest remaining width is 0.00013796, below 10^{-3} . Eighteen cases have strictly smaller upper bounds and strictly smaller gaps than the harmonic comparator; five are equal and one has a slightly wider interval, still below the requested tolerance. A stronger exact model does not require every interrupted search bound to dominate a differently partitioned relaxation. The reversal is retained as case 19. Table 2 reports every case. The 12 small-case exact optima remain inside the new intervals. These results test the stronger model directly: they do not subtract an estimated dispersion error from a reduced solver’s gap.

Many distinct histories with common eligibility. The next 16 instances have 16, 64, 256, or 1,024 histories, catalog sizes nine or seventeen, budget three, two declared seeds, nonuniform weights, continuously varying rational caps, positive opening charges, and curvatures including zero. The root obligation is nine-tenths of aggregate capacity. All have a single eligible catalog prefix. The exact algorithm closes every interval in one node with tolerance zero. Table 3 reports optimization and independent verification times

separately. The polynomial statement is Theorem 4.2; these finite measurements illustrate its behavior and do not establish a machine-independent runtime law.

Eligibility complexity remains consequential. Six additional 12-history instances have five to eight eligibility classes. Each is run at tolerances 10^{-2} and 10^{-3} and node limits 31 and 127, giving 24 specified runs. All twelve requests at 10^{-2} finish; at 10^{-3} , two of twelve finish. Thus ten tighter requests remain unresolved, and every one is retained. At 127 nodes the unresolved 10^{-3} gaps range from about 0.0012862 to 0.0059058. Table 4 gives both node budgets and exact exhaustive references; the full record also includes the easier tolerance. Increasing the node allowance weakly tightens these observed bounds, but the fixed budget does not remove the theoretical exponential dependence on eligibility complexity.

Independent formulations and semantic checks. The direct uneliminated formulation uses activation and support binaries, service tiers, branch lotteries, the exact promise, and supportwise realization ceilings. Tangent and secant quadratic envelopes are solved for all 12 primary paired cases and all six stress instances, with 32 segments and three seconds per solve. All 36 solves terminate with the solver’s success status in this execution. None of the finite envelope brackets is relabeled as an exact rational certificate. The computational record includes every solver time, node count, relative gap, envelope error, and bracket width. Exact reference values lie inside all available numerical brackets, allowing the stated numerical tolerance.

All 64 original-space certificates pass a separate checker. Exact regressions compare 564 grouped price bounds with complete book enumeration, verify 72 recovered fixed-mean allocations, ten common-eligibility global optima, eleven boundary global cases, and five multigroup intervals. Sixteen deliberately corrupted certificates are rejected, including errors in service prices, costs, corrections, anchors, budgets, and the target cover. These checks complement the proofs: agreement of two numerical implementations is not used in place of a coverage or weak-duality argument.

10 Conclusion

The relevant dimension of joint renewal design is not necessarily the number of observed histories. Common catalog eligibility permits exact pooling of arbitrary participation caps and quadratic service costs. Terminal-first allocation isolates the book dependence of heterogeneous service in a last-eligible-level correction. Paying that correction on the ordered path gives a polynomial exact joint algorithm for one eligibility class and an exact group-box price oracle for several classes.

The resulting certificates retain the original command budget, opening charges, aggregate promise, and individual constraints. Their adaptive accuracy dependence is controlled by eligibility complexity rather than history count or dispersion bins. Exact single-class computations and checked multigroup intervals are reported separately; the latter include the unsuccessful accuracy requests. The preserved harmonic, continuous, institutional, Monge, compression, and inexact-support results describe additional regimes rather than substitute for the new exact joint theorem. The study establishes algorithmic and certificate behavior on specified synthetic inputs, not a calibrated field effect.

Code and data availability. The accompanying repository contains the immutable review and scientific-parent identifiers, the pre-execution protocol, all rational instances and policy certificates, independent checkers, numerical mixed-integer records, source files, and build instructions. The computational record identifies every case, including interruptions, and separates historical, local-validation, and publication-run timing environments. A single certificate can be checked without rerunning any optimizer. The terminal alphabet count is not a claim about total implementation memory: target tables, encoder logic, read-only constants, probability precision, and certificate storage remain distinct resources.

A Joint Approximation Without Additional Memory

The exact conditional oracle permits a global approximation of the original catalog problem without enumerating books or increasing their cardinality. The price is dependence on the dimension of the target vector. Let f be L_f -Lipschitz and each h_j be $L_{h,j}$ -Lipschitz on $[0, b_j]$, and set $L = \max\{L_f, \max_j L_{h,j}\}$. For the quadratic implementation, $L = \max\{r, \max_j \gamma_j\}$ is valid. These assumptions bound value changes, not the sensitivity of a selected alphabet.

Lemma A.1 (Target sensitivity and exact repair). *For every feasible book c , its branch response is L -Lipschitz on $[c_1, b_j]$. Given $u_j \in [c_1, b_j]$ and a feasible root promise B , bounded transfers produce $t_j \in [c_1, b_j]$ with $\sum_j \pi_j t_j = B$ and*

$$\sum_j \pi_j |t_j - u_j| = \left| B - \sum_j \pi_j u_j \right|. \quad (42)$$

The operating payoff decreases by at most L times the right-hand side. With rational inputs, both transfers and canonical lotteries are rational and satisfy the original constraints exactly.

Proof. Chord slopes of f have absolute value at most L_f ; the final shortfall tail has slopes bounded by $L_{h,j}$. Their continuous concatenation is therefore L -Lipschitz. If the residual is positive, increase targets within their remaining cap capacities; if it is negative, decrease them toward c_1 . Total available capacity is sufficient because $B \in [c_1, \bar{b}]$. Stop when the residual is zero. All changes have the same sign, proving (42); the payoff bound follows by the weighted Lipschitz inequality. Reconstruct the lotteries after repairing targets, rather than rounding their probabilities separately. The canonical rule then enforces $y_j + C_j \leq \tau_j$ for each realization. $\square$

Theorem A.2 (Same-budget additive scheme at fixed branch count). *For a feasible finite catalog and any $\varepsilon > 0$, a target-net algorithm returns one feasible book of at most m levels and net value J satisfying*

$$\mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho) - \varepsilon \leq J \leq \mathcal{V}_{m,\mathcal{A}}^\tau(B; \rho). \quad (43)$$

For $k \geq 2$, set $\eta = \varepsilon / [2L(k-1)]$. At most

$$N(1 + 1/\eta)^{k-1} \quad (44)$$

target profiles need be optimized. Thus a safe arithmetic bound is $O((k+m)N^3(1+2L(k-1)/\varepsilon)^{k-1})$. For $k = 1$, apply the conditional oracle once at $t_1 = B$. The guarantee allows arbitrary nonnegative catalog charges and arbitrary ceilings $\tau_j \geq b_j$; neither is approximated. For rational quadratic data the algorithm is polynomial in input bit length and the displayed numerical accuracy parameter when k is fixed. It is not an FPTAS with variable k and does not claim polynomial dependence on the binary length of $1/\varepsilon$.

Proof. Enumerate a possible lowest level $a \in \mathcal{A}$ with $a \leq a_* := \min\{B, \min_j b_j\}$. Restrict the conditional candidate pool to levels at least a . In weighted target coordinates, let $x_j = \pi_j(t_j - a)$, $d_j = \pi_j(b_j - a)$, and $C = B - a$. Choose one distinguished branch, numbered k only for this proof. Enumerate multiples of η in $[0, d_j]$ for $j < k$, discarding tuples whose sum exceeds C . Set

$$x_k = \min\{d_k, C - \sum_{j < k} x_j\}.$$

Any remaining deficit is nonnegative; fill it by bounded increases to the coordinates $j < k$. Their total available capacity is sufficient because $C \leq \sum_j d_j$. This constructs a feasible profile $t_j = a + x_j/\pi_j$. Run Theorem 3.1 at budget m and retain the best policy over all anchors and profiles.

Let (c^*, t^*) be a global catalog optimum, which exists by compactness of each fixed-book allocation problem and finiteness of the catalog. Its lowest level $a = c_1^*$ is enumerated. For $j < k$, the enumeration includes $x_j = \eta \lfloor x_j^*/\eta \rfloor$. Their aggregate decrease is $D < (k-1)\eta$. The constructed last coordinate is at least x_k^* and increases by at most D . If it reaches its cap, the subsequent increases on the other coordinates use only the remaining deficit. Consequently, the final weighted target distance from t^* is at most $2D \leq 2(k-1)\eta$. All final targets remain at least a , so the *same* book c^* is feasible. Lemma A.1 bounds its payoff loss by ε ; its opening charge is unchanged. The conditional optimizer does no worse. Maximizing over profiles proves (43).

Each of the $k-1$ coordinates has at most $1 + d_j/\eta \leq 1 + 1/\eta$ possibilities. There are at most N anchors. Applying the conditional-oracle complexity gives the stated bound. The construction uses rational flooring, addition, comparison, and the rational oracle, so the bit-complexity conclusion follows under the stated representation assumptions. $\square$

The scheme is useful when the number of distinct contract branches is small or repeated homogeneous branches have been aggregated. Its exponential accuracy dependence on k is not concealed by the polynomial dependence on the symbol budget. At a large branch count we use selected target profiles and certified upper bounds rather than declare the net completed. An interrupted enumeration returns a feasible incumbent, but the implementation deliberately withholds the requested ε guarantee until coverage is complete.

This result addresses the original-budget difficulty differently from union recovery. It pays for accuracy in target-space computation, not extra installed levels. At saturation, Corollary 3.2 avoids target-space enumeration altogether. For a general interior promise, selected profiles can be polished by alternating exact inner allocation and exact conditional book design. This preserves feasibility and nondecreases value at every step, but a fixed iteration cap or stationary pair has no global-optimality implication. The comparative study records this distinction explicitly.

B Eligibility-Prefix Decomposition

The closed-cell formulation establishes exact continuous attainability. We now exploit the ordered interface to avoid enumerating either branch modes or catalog subsets in a price subproblem. The resulting bounds support finite global search and remain valid when that search stops early. This distinction matters: a price relaxation is not, without an additional equality, the original optimum.

B.1 A reusable ordered-interface allocation class

Let $\mathcal{A} = (a_1 < \dots < a_N) \subset [0, 1]$ be a catalog with nonnegative opening charges ρ_i . The common reward f is strictly increasing and concave. Branch j has weight π_j , target cap b_j , nondecreasing convex shortfall cost h_j with $h_j(0) = 0$, and an eligibility threshold $\tau_j \geq b_j$. Its response is (2) with eligible levels $c \cap [0, \tau_j]$. The renewal model has $\tau_j = b_j + \delta$; the theorem below permits distinct thresholds without requiring a common overrun tolerance. This abstract class needs no acceptance or renewal interpretation. It describes any ordered shared interface with a common concave output reward, separable shortfall costs, and these target and support restrictions.

Write $a_* = \min\{B, \min_j b_j\}$. Assume $a_1 \leq a_*$, since otherwise no feasible book exists, and replace m by $\min\{m, N\}$. Restrict books to $c_1 \leq a_*$; this retains exactly those books that can meet the root promise. At a real price λ , define

$$g_\lambda(x) = f(x) - \lambda x, \quad \phi_j(z; \lambda) = \max_{0 \leq y \leq z} \{-h_j(y) - \lambda y\}, \quad (45)$$

$$D(\lambda) = \lambda B + \max_{\emptyset \neq c \subset \mathcal{A}, |c| \leq m, c_1 \leq a_*} \left[\sum_j \pi_j \max_{c_1 \leq t \leq b_j} \{v_{j,c}(t) - \lambda t\} - \sum_{a_i \in c} \rho_i \right]. \quad (46)$$

Weak duality gives $\mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq D(\lambda)$ in the renewal specialization. For quadratic shortfall costs, (45) is evaluated at $y = 0$ if $\lambda \geq 0$, and at $y = \min\{z, -\lambda/\gamma_j\}$ otherwise; when $\gamma_j = 0$ and $\lambda < 0$, use $y = z$.

Call an arc $u < v$ admissible when $g_\lambda(v) \geq g_\lambda(u)$. Define its reward and the reward for stopping at u by

$$E_\lambda(u, v) = \sum_{j: u \leq b_j < v} \pi_j \begin{cases} g_\lambda(u) + \frac{b_j - u}{v - u} [g_\lambda(v) - g_\lambda(u)], & v \leq \tau_j, \\ g_\lambda(u) + \phi_j(b_j - u; \lambda), & v > \tau_j, \end{cases} \quad (47)$$

$$T_\lambda(u) = \sum_{j: b_j \geq u} \pi_j [g_\lambda(u) + \phi_j(b_j - u; \lambda)]. \quad (48)$$

An edge accounts only for branches whose target caps it crosses. It therefore carries a contractual allocation value, not an ordinary nearest-neighbor quantization error.

Theorem B.1 (Polynomial price decomposition). *For every λ , some maximizer in (46) has nondecreasing g_λ along its ordered book. Its priced operating value is the sum of (47) over consecutive levels plus (48) at*

the last level. Set

$$W_1(i) = T_\lambda(a_i), \tag{49}$$

$$W_r(i) = \max \left\{ T_\lambda(a_i), \max_{\substack{\ell > i \\ g_\lambda(a_\ell) \geq g_\lambda(a_i)}} [E_\lambda(a_i, a_\ell) - \rho_\ell + W_{r-1}(\ell)] \right\}.$$

Then $D(\lambda) = \lambda B + \max_{a_i \leq a_s} [W_m(i) - \rho_i]$. With constant-time evaluation of ϕ_j , computing a maximizing book and this bound takes $O((k+m)N^2)$ arithmetic operations and $O(N^2 + mN)$ storage, allowing both k and m to vary. Rational quadratic data and a rational price give an exact rational algorithm with polynomial bit complexity. General costs use at most $O(kN^2)$ calls to the scalar oracle (45) instead.

Proof. If $\lambda < 0$, strict increase of f makes g_λ strictly increasing. If $\lambda \geq 0$, intermediate service cannot improve priced payoff, and a branch maximizes the interpolant of g_λ on its eligible prefix, truncated at its target cap. Concavity implies that the values along any book increase up to their first maximum and never subsequently increase. All levels after that first maximum may be deleted: for a branch below the maximizing level, either that level is ineligible and all later levels are also ineligible, or later chords cannot improve the concave interpolant before the cap; for a branch above it, no later value beats the maximum. Nonnegative charges make deletion weakly beneficial. The retained book has only admissible arcs.

Consider $b_j \in [u, v)$ on such a book. If $v \leq \tau_j$, the best priced target is b_j on the chord $[u, v]$. For $\lambda \geq 0$ its slope is nonnegative and a singleton tail contributes no positive adjustment. For $\lambda < 0$, a singleton tail contributes at most $-\lambda(b_j - u)$, whereas the chord contributes this amount plus a strictly positive reward increment when $b_j > u$. If $v > \tau_j$, every later level is ineligible; the optimum is a singleton at u followed by the best shortfall in (45). Branches beyond the final level have the same singleton-tail form. These disjoint cap intervals cover all branches because the first level is at most $\min_j b_j$. This proves the decomposition, including equality cases.

The acyclic recurrence chooses either to stop or to append the next admissible level, paying each selected charge once. It enumerates paths implicitly rather than enumerating books. Computing edges costs $O(kN^2)$; the dynamic-programming layers cost $O(mN^2)$. For rational quadratic data, every scalar oracle value uses a bounded number of rational operations, and each path value sums at most $O(km)$ such terms. Numerator and denominator lengths are polynomial in the input bit length; comparisons and path reconstruction therefore preserve polynomial bit complexity. $\square$

Pricing and constrained shortest-path bounds are established optimization mechanisms (Handler and Zang, 1980). The model-specific result here is that one price makes an otherwise nonseparable shared-book allocation decompose into these cap-crossing edges, despite heterogeneous shortfall costs, branch-specific eligibility, and nonsaturated targets. The theorem does not assert a polynomial algorithm for the unrelaxed joint problem or exchange its maximization with minimization over prices.

B.2 Prefix bounds and checkable global search

A node is a mandatory initial segment $p = (i_1, \dots, i_s)$ of a book; its descendants may append only larger indices. If the priced values along p never decrease, its exact priced completion value is

$$U_p(\lambda) = \lambda B + \sum_{h=1}^{s-1} E_\lambda(a_{i_h}, a_{i_{h+1}}) - \sum_{h=1}^s \rho_{i_h} + W_{m-s+1}(i_s). \quad (50)$$

If a decrease occurs in p , concavity places its first priced maximum within p and every larger new level is useless to the priced allocation. In that case, use the fixed-prefix branch-support value, with every mandatory charge retained. This is again the exact priced maximum over completions; additional nonnegative charges cannot help. The empty prefix uses (46).

Theorem B.2 (Finite search and an anytime global interval). *Fix any nonempty finite set of prices. At each prefix use the smallest corresponding $U_p(\lambda)$, evaluate feasible books by the exact inner allocator, and retain the largest feasible net value L . Partition an unpruned node into its own book, when nonempty, and all one-index extensions. Prune a node only when its upper bound is at most L . This procedure terminates after finitely many prefixes and returns the exact catalog optimum. At any interruption it returns*

$$L \leq \mathcal{V}_{m,\mathcal{A}}^\delta(B; \rho) \leq U = \max\{L, \max_{p \text{ unresolved}} \min_{\lambda} U_p(\lambda)\}. \quad (51)$$

For a uniform mesh of spacing h , Theorem F.1 further yields the continuous interval $[L, U + K_m h]$. The optimization gap $U - L$ and discretization error $K_m h$ are separate and observable.

Proof. The preceding completion argument proves each price bound, so their minimum is valid by weak duality. Every feasible book either equals a nonempty node or has a unique next index. Hence the prescribed children and the node's own book are a disjoint exhaustive partition; book size is at most m . A pruned subtree cannot improve the incumbent. The maximum of unresolved upper bounds and the incumbent therefore bounds every feasible book at every stage. If all subtrees have been resolved, the two bounds coincide. The mesh transport is a separate one-sided inequality between the continuous optimum and the exact catalog optimum, so substituting the catalog upper bound proves the last claim. $\square$

A global witness consists of the feasible winning lottery, every split and its complete child list, upper bounds and prices at pruned or unresolved nodes, and a fixed-book supporting price for each explicitly evaluated node book. A checker can rebuild price tables and verify the partition without replaying the search order or trusting the incumbent-selection code. This is a stronger object than an inner multiplier or a source hash. Equality of one root price bound with a feasible payoff already certifies global catalog optimality without any branching. Worst-case search is still exponential in the alphabet budget; neither this result nor the mesh transfer is an FPTAS claim.

Additional decomposition contribution. Theorems B.1–B.2 use classical Lagrangian bounds and an acyclic path recurrence. Their structural increment is the exact eligibility-aware cap-crossing reward and

prefix-completion dominance for nonsaturated allocation with heterogeneous shortfall costs. The generic search framework is not claimed as new. Proposition F.3 establishes sharpness for uniform catalogs, not a lower bound for adaptive grids. The continuous stationary-face mechanism remains generic; its contractual closed-cell reduction remains the structural contribution.

C Exact Continuous Quadratic Design

Fix $s \leq m$ and temporarily allow $0 \leq c_1 \leq \dots \leq c_s \leq 1$. For each branch select one of $2s - 1$ modes. A singleton mode S_ℓ and an adjacent-lottery mode L_ℓ have the following closed descriptions:

$$S_\ell : \quad c_\ell \leq t_j \leq b_j, \quad g_j^S = rc_\ell - \frac{q}{2}c_\ell^2 - \frac{\gamma_j}{2}(t_j - c_\ell)^2; \quad (52)$$

$$L_\ell : \quad c_\ell \leq t_j \leq c_{\ell+1}, \quad t_j \leq b_j, \quad c_{\ell+1} \leq b_j + \delta, \quad g_j^L = rt_j - \frac{q}{2}[(c_\ell + c_{\ell+1})t_j - c_\ell c_{\ell+1}]. \quad (53)$$

For S_ℓ , use $y_j = t_j - c_\ell$ and terminal level c_ℓ . For L_ℓ , use $y_j = 0$ and the adjacent unbiased lottery. At $c_\ell = c_{\ell+1}$ the constraints force $t_j = c_\ell$, and the formula is the singleton reward. Therefore collisions are not discarded as singular probability formulas. The quadratic interpolation identity eliminates the ratio in the lottery probabilities before codewords become optimization variables.

Each joint mode assignment, the ordered-book inequalities, and $\sum_j \pi_j t_j = B$ define a compact polytope. If $\rho(c) = \alpha c^2 + \beta c + \eta$ is nonnegative on $[0, 1]$, its objective

$$Q(c, t) = \sum_j \pi_j g_j(c, t) - \sum_{\ell=1}^s \rho(c_\ell) \quad (54)$$

is a quadratic polynomial, not necessarily concave. Nonnegative charges, rather than signs of their coefficients, are the assumption. For example, a decreasing nonnegative certification charge is allowed.

Lemma C.1 (Finite stationary-face search). *A quadratic function on a nonempty compact polytope attains its maximum at a point that is either a vertex or the unique stationary point on the affine hull of some face with negative-definite restricted Hessian. With rational input data, some maximizing point and its value are rational and can be found by finitely many rational linear solves and feasibility tests.*

Proof. Among all faces whose relative interiors contain a global maximizer, choose one of least dimension and a maximizing point x in its relative interior. The derivative along the face's tangent space vanishes and the restricted Hessian is negative semidefinite. If this restriction has a nonzero null direction z , the quadratic is constant along $x + az$ in the affine hull: both its linear and quadratic terms in a vanish. Compactness makes this line reach a proper face while remaining feasible, contradicting the choice of dimension. The restriction is therefore negative definite unless the tangent space is zero dimensional.

Enumerate independent subsets of the defining inequalities, together with the equality constraints, and solve the stationarity equations on each associated affine space. On the face just identified the bordered stationarity matrix is nonsingular; at a vertex, the tangent space is empty and the same conclusion holds. Test feasibility in every original inequality and retain the largest objective among all feasible candidates.

Each retained candidate is feasible, and at least one is a global maximizer. Rational elimination gives the rationality claim. Singular bordered systems may be skipped because the argument guarantees a nonsingular maximizing face of smaller dimension; it does not assume nondegeneracy of the original polytope. $\square$

Theorem C.2 (Exact joint continuous design). *For rational quadratic primitives, rational caps and probabilities, rational $B \in [0, \bar{b}]$ and $\delta \geq 0$, and a rational quadratic charge nonnegative on $[0, 1]$, the maximum of (54) over the closed cells (52)–(53), for every $1 \leq s \leq m$, equals $\mathcal{V}_m^\delta(B; \rho)$. The optimum is attained by a rational book, rational targets, and rational lottery probabilities. Deleting the lottery modes gives the exact deterministic frontier at every promise, with the same charges.*

An exact algorithm enumerates at most $(2s - 1)^k$ cells at size s . After eliminating one target, each cell has dimension at most $s + k - 1$ and at most $s + 1 + 4k$ inequalities. Enumerating its independent active sets and applying Lemma C.1 gives a finite global algorithm with at most

$$\sum_{s=1}^m (2s - 1)^k 2^{s+1+4k} \quad (55)$$

linear systems, each of polynomial dimension and rational encoding length in the input size. The sharper rank-limited count replaces 2^{s+1+4k} by $\sum_{r=0}^{\min\{s+k-1, s+1+4k\}} \binom{s+1+4k}{r}$. A lottery branch may impose four separate inequalities: its two chord bounds, its target cap, and its realization eligibility bound. Neither of the latter two is redundant at a general positive tolerance. This is an exponential reference algorithm, not a polynomial-time claim or an extension of the saturated Monge complexity.

Proof. Theorem 2.1 turns every optimal book and target allocation into singleton or adjacent-lottery branch modes represented in these cells. Conversely every cell point implements a feasible policy by the displayed reconstruction; all constraints, including the exact root promise, hold. If ordered slots collide or some slots are unused, deleting redundant slots cannot increase a nonnegative charge. The resulting smaller book and its canonical branch rules appear among the enumerated sizes. Hence overcounting charges at duplicate slots cannot change the maximum across all sizes. These two directions prove equality with the original optimization problem, not merely a relaxation or a fixed-book upper bound.

There are finitely many compact cells and each has a continuous quadratic objective. Lemma C.1 supplies a rational maximizing candidate on each nonempty cell. Adjacent distinct rational levels reconstruct rational probabilities; coincident levels reconstruct a singleton. The mode count, inequality count, and all-subsets upper bound give (55). Standard determinant bounds for rational linear systems give polynomial encoding length for each candidate. Singleton-only modes exactly represent deterministic terminal assignments and intermediate tiers, proving the deterministic statement. $\square$

The reduction solves the unrestricted continuous outer problem at an interior promise, rather than relabeling fixed-book allocation as global design. Its contract-specific content is the eligibility-prefix reduction and the closed quadratic elimination of lottery probabilities with no loss at collisions. Rational quadratic optimization and active-set reasoning are classical; polynomial-size witnesses are not a new general complexity principle (Del Pia et al., 2017). The fast saturated Monge theorem remains valuable precisely because the general finite search is exponential.

C.1 Pathwise comparison, repeated caps, and small charges

Theorem C.2 computes $V_m^E(B)$, $V_m^P(B)$, and $V_m^D(B)$ for every rational promise, using respectively all lottery modes at $\delta = 1$, all modes at $\delta = 0$, and singleton modes. No claim of pathwise–deterministic equality away from saturation is needed for this characterization. The complete predecessor companion retains the earlier exact institutional comparison path; the current study reports the separately executed budgeted–design comparisons. The zero-memory-loss interval $B \leq \min b_j$ in Theorem EC.4.1 remains important: a positive institution gap is not expected throughout the feasible promise range.

Proposition C.3 (Pathwise randomization can matter in the interior). *For one branch with $b = 9/10$, $B = 9/20$, $r = 2$, $q = \gamma = 1$, two available symbols, and the nonnegative charge $\rho(c) = 3c(1 - c)$,*

$$\mathcal{V}_2^0(B; \rho) = \frac{171}{400} > \mathcal{V}_2^D(B; \rho) = \frac{9}{160}.$$

The pathwise optimum uses book $(0, 9/10)$, equal draw probabilities, and no intermediate tier. The deterministic optimum uses the single level $9/20$.

Proof. A deterministic policy uses only one level $0 \leq c \leq B$; deleting unused levels weakly lowers its charge. Its net payoff is $-81/800 - (11/20)c + 2c^2$, whose maximum over $[0, B]$ is $9/160$ at B . Every nonsingleton canonical pathwise mode has $0 \leq u \leq B \leq v \leq b$, and net payoff $2B - (u+v)B/2 + uv/2 - 3u(1-u) - 3v(1-v)$. This is convex in (u, v) , so its maximum on that rectangle is at a corner. Checking its four corners, together with the singleton case, gives $(u, v) = (0, 9/10)$ and value $171/400$. Both realizations meet the hard cap and their mean is exactly B . $\square$

This example uses a specified nonnegative, nonmonotone level charge; it is not a counterexample asserted for the uncharged problem. It shows why the saturated pathwise collapse cannot simply be imported into the unified interior design model.

Proposition C.4 (Homogeneous repeated-cap aggregation). *For the randomized institutions in (1), branches with identical caps and identical intermediate-cost functions can be aggregated by adding their probabilities, at every B , δ , and codeword charge. This holds for each fixed book and after global book optimization.*

Proof. Such branches have identical eligible prefixes and the same concave response (2). Replacing their targets by their probability-weighted mean preserves the root promise and does not lower total reward by Jensen’s inequality. Replicating an aggregate target gives the converse inequality. Charges depend only on the common book. This argument does not aggregate heterogeneous costs or establish a corresponding deterministic restriction. $\square$

The strict off-cap conclusion is also stable under genuinely selected-level charges. In Proposition D.3, the uncharged two-level book $(1/4, 5/8)$ beats every book of at most two cap levels by $1/1280$. If $0 \leq \rho(c) \leq \eta$ on $[0, 1]$ and $\eta < 1/2560$, evaluating this same off-cap book incurs less than that gap, while a cap-only book cannot improve its uncharged payoff after a nonnegative charge. Thus no charged optimum is cap-only. This statement preserves the off-cap property, not necessarily the numerical location $5/8$.

D The Global Continuous Lottery Codebook

The fixed-codebook formula in Theorem EC.3.1 leaves a continuous design problem: both the terminal service levels and their assignment probabilities must be chosen. We now solve that problem globally for quadratic terminal rewards and heterogeneous quadratic intermediate costs. The solution also determines when imposing participation after, rather than before, the private draw eliminates the value of randomization. These are restricted-alphabet design results, distinct from the exact-implementation alphabet in Theorem EC.4.1.

Retain the three-date renewal architecture of Theorem EC.2.1, including the saturated root promise $\bar{b} = \sum_{j=1}^k \pi_j b_j$, strictly ordered caps $0 < b_1 < \dots < b_k < 1$, and positive probabilities summing to one. Throughout this section,

$$f(c) = rc - \frac{q}{2}c^2, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad r \geq q > 0, \quad \gamma_j \geq 0. \quad (56)$$

The parameters and probabilities can differ across branches as indicated. Terminal reward is strictly increasing on the feasible cap range, including when $r = q$. All private information retained until the common terminal vertex must be encoded in the charged symbol. The decoder and public observation convention are unchanged.

D.1 Cap anchors and one continuous terminal level

For $u \leq b \leq v$, quadratic concavity gives the interpolation loss

$$f(b) - \frac{v-b}{v-u}f(u) - \frac{b-u}{v-u}f(v) = \frac{q}{2}(b-u)(v-b), \quad (57)$$

with the endpoint value understood when levels coincide. Thus an interior codeword enters the loss affinely until it crosses a branch cap. The highest codeword is different: reducing it below a branch cap creates an intermediate payment and its associated cost.

Lemma D.1 (Cap-anchored representation). *For every alphabet budget $m \geq 1$, an optimum of (EC.5) exists with the following representation. Its lowest codeword is b_1 ; every codeword except the highest belongs to $\{b_1, \dots, b_k\}$; and the highest belongs to $[b_1, b_k]$. A one-symbol optimum is $\{b_1\}$. The highest codeword need not be a branch cap.*

The proof is in Electronic Companion Section EC.5.4.

The lemma does not discretize the highest level. A cap-only restriction can be strictly suboptimal, as Proposition D.3 shows. It also does not assert that every optimal codebook is anchored; it supplies an optimum with this useful form.

D.2 A finite exact algorithm

For adjacent anchored levels $b_i < b_j$, define

$$E(i, j) = \frac{q}{2} \sum_{h=i}^j \pi_h (b_h - b_i)(b_j - b_h). \quad (58)$$

Let $P(s, j)$ be the minimum interpolation loss on caps through b_j using exactly s anchored levels, starting at b_1 and ending at b_j . Set $P(1, 1) = 0$, $P(1, j) = +\infty$ for $j > 1$, and use the ordered recurrence

$$P(s, j) = \min_{i < j} \{P(s-1, i) + E(i, j)\}, \quad 2 \leq s \leq j. \quad (59)$$

This is a standard shortest-path or segmentation recurrence. The substantive reduction is Lemma D.1, which makes it applicable to a jointly optimized continuous lottery codebook.

Suppose that b_i is the last anchored level and z is the final level. For $\ell \geq i$ and $b_\ell \leq z \leq b_{\ell+1}$, put $b_{k+1} = b_k$ and define

$$\begin{aligned} \Psi_{i\ell}(z) = & \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)(z - b_h) \\ & + \sum_{h=\ell+1}^k \pi_h \left\{ f(b_h) - f(z) + \frac{\gamma_h}{2} (b_h - z)^2 \right\}. \end{aligned} \quad (60)$$

The first sum is the lottery loss below z ; the second is the terminal-reward and intermediate-payment loss above it. Empty sums are zero. For $\ell < k$ the function is strictly convex in z and has the constrained minimizer

$$z_{i\ell} = \left[\frac{r \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h b_h - \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)}{q \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h} \right]_{[b_\ell, b_{\ell+1}]}. \quad (61)$$

For $\ell = k$, take $z_{ik} = b_k$. Define the terminal cost

$$T_i = \min_{i \leq \ell \leq k} \Psi_{i\ell}(z_{i\ell}). \quad (62)$$

Theorem D.2 (Global frontier and exact complexity). *Under (56), the optimum with at most m symbols is*

$$D_m^{\text{ran}} = \min \left\{ D_1, \min_{\substack{1 \leq s \leq \min(m-1, k) \\ 1 \leq j \leq k}} \{P(s, j) + T_j\} \right\}, \quad (63)$$

where the inner minimum is empty for $m = 1$. Backtracking produces a globally optimal codebook and the adjacent lotteries implementing it. For ordered inputs and $1 \leq m \leq k$, all frontiers through budget m can be computed in $O(mk^2)$ exact rational operations and $O(mk + k)$ stored rational numbers and indices, including backpointers. No codeword grid is required.

For rational input of total binary length L , a reduced-rational implementation has polynomial Turing

complexity. A conservative bound is $O(mk^2S^3)$ bit operations with $S = O(k[L + \log(k+2)])$. This includes exact cost comparisons, clamping, and fraction reduction. For $m \geq k$ the optimal loss is zero; listing additional budget entries requires only their output cost. These guarantees concern the stated algorithm, not an arbitrary nonlinear optimizer for (EC.5).

The proof is in Electronic Companion Section EC.5.5.

D.3 A continuous optimal highest level

Proposition D.3 (An off-cap terminal level is optimal). *Let $(b_1, b_2, b_3) = (1/4, 1/2, 3/4)$, $(\pi_1, \pi_2, \pi_3) = (7/20, 3/5, 1/20)$, $r = 2$, $q = 1$, and $\gamma_j = 1$. The globally optimal two-symbol expected-participation codebook is $(1/4, 5/8)$, and*

$$D_2^{\text{ran}} = \frac{23}{1280} < \frac{3}{160} = \min_{\substack{|c| \leq 2 \\ c \subseteq \{b_1, b_2, b_3\}}} D(c) = D_2, \quad (64)$$

where the cap-only minimum is over feasible codebooks and $D(c)$ denotes their expected-participation loss. Thus even optimizing all cap-only lotteries misses the unrestricted optimum.

The proof is in Electronic Companion Section EC.5.6.

D.4 Participation after the private draw

The preceding frontier is appropriate when a renewal review approves an expected continuation before the protocol draws its private symbol. Consider instead an agreement that requires the same cap to hold for every private realization after that review. A draw-specific intermediate tier is allowed; only the terminal public state and the charged symbol reach the terminal decoder. Write $D_m^{\text{pw,ran}}$ for the best randomized loss under these pathwise constraints and the same saturated root promise.

Theorem D.4 (Pathwise randomization has no frontier advantage). *For the general renewal family of Theorem EC.2.1, with strictly increasing concave f and convex nondecreasing h_j , and the observation architecture just specified,*

$$D_m^{\text{pw,ran}} = D_m \quad \text{for every } m \geq 1. \quad (65)$$

Consequently (63) and the deterministic frontier give an exact comparison of the two participation institutions under (56). Their exact-optimum alphabet thresholds are both k when strict concavity makes the k distinct full-information actions unique.

The proof is in Electronic Companion Section EC.5.7.

This institutional result is not an assertion that randomization is generally ineffective in dynamic contracts. It uses the saturated root promise, an exogenous recombination, and the charged-information interface. Under expected participation, Theorem D.2 chooses a less expensive lottery protocol; under pathwise participation, the same service model calls for the deterministic frontier. In either institution an operator can price a symbol budget, add its implementation charge to the computed loss, and optimize the budget without confusing read-only program size with retained private information.

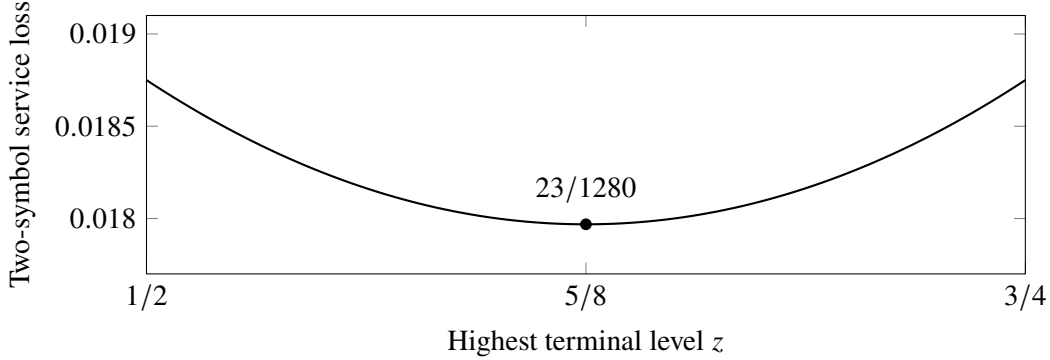


Figure 1: The analytically optimized highest level in the heterogeneous three-branch example. The lower level is $1/4$. The displayed function is $z^2/20 - z/16 + 3/80$ on $[1/2, 3/4]$; its interior minimum is $z = 5/8$, not a branch cap.

E Monge Costs and Classical Linear Matrix Search

The finite dynamic program still optimizes a continuous highest service level. We first prove the contractual cost inequalities and then verify the classical easy-direction completion with the exact order used by the code. A search routine cannot establish the economic reduction that supplies its entries.

E.1 The inequality survives continuous terminal minimization

Write $I_\ell = [b_\ell, b_{\ell+1}]$, including $I_k = \{b_k\}$, and $Q(i, \ell) = \min_{z \in I_\ell} \Psi_{i\ell}(z)$ for $i \leq \ell$. A partial array is Monge here when

$$A(r, c) + A(s, d) \leq A(r, d) + A(s, c), \quad r < s, \quad c < d, \quad (66)$$

whenever all four entries are defined. This fixes the inequality convention before any orientation is named.

Lemma E.1 (Contractual finite-cost inequalities). *The anchored edge costs E and deterministic cell costs C satisfy $A(a, c) + A(b, d) \leq A(a, d) + A(b, c)$ for $a \leq b \leq c \leq d$. The continuously minimized terminal array Q is Monge on its feasible domain. These statements permit arbitrary positive probabilities and heterogeneous nonnegative intermediate curvatures.*

The proof is in Electronic Companion Section EC.5.8.

Lemma E.2 (Previous-layer values are column offsets). *Adding any finite column term v_c preserves the Monge inequality. For fixed finite predecessor values, the arrays $A(j, i) = P(s, i) + E(i, j)$ and $B(j, i) = D(s - 1, i) + C(i + 1, j)$ are therefore Monge wherever all four entries are feasible.*

The proof is in Electronic Companion Section EC.5.9.

E.2 Completion with the implemented strict and tie orders

Give a feasible entry the lexicographic key

$$K(r, c) = (0, A(r, c), c). \quad (67)$$

Finite ties select the smaller column, and distinct columns have distinct keys. The exact total-monotonicity condition used by the implementation is

$$K(r, d) < K(r, c) \implies K(s, d) < K(s, c), \quad r < s, c < d. \quad (68)$$

A strict preference for the right column cannot reverse in a later row. This applies to arbitrary selected rows and columns. Because keys are distinct, a decrease of row-argmin indices in any submatrix would contradict (68).

Lemma E.3 (Suffix-feasible completion). *Let row r have feasible columns $c \geq \ell_r$, with nondecreasing cutoffs ℓ_r . Suppose the finite entries are Monge. Complete every $c < \ell_r$ with $K(r, c) = (1, 0, -c)$. Then (68) holds. A completely infeasible row of a selected submatrix selects its largest selected column.*

The proof is in Electronic Companion Section EC.5.10.

Lemma E.4 (Prefix-feasible completion). *Let row r have feasible columns $c \leq u_r$, with nondecreasing cutoffs u_r . Suppose the finite entries are Monge. Complete every $c > u_r$ with $K(r, c) = (1, 0, c)$. Then (68) holds. A completely infeasible selected row selects its smallest selected column.*

The proof is in Electronic Companion Section EC.5.11.

These lemmas specialize the known favorable completion setting in Section 1; they are not general partial-matrix search theorems. A helper accepting two bounds does not prove total monotonicity for arbitrary interval rows. For example, two zero-cost rows feasible only at columns 2 and 1 have a vacuously Monge finite part but decreasing minima. Uniform infinite padding with ordinary left ties also fails on the required suffix domain: rows with cutoffs 2, 3 restricted to columns 1, 2 select 2, 1. The opposite internal padding orders above are deliberate.

E.3 Reachability and three search interfaces

Lemma E.5 (Contiguous reachable predecessor sets). *For the anchored program, $P(1, j)$ is finite exactly at $j = 1$, and for $s \geq 2$, $P(s, j)$ is finite exactly when $s \leq j \leq k$. For the deterministic program, $D(0, j)$ is finite only at $j = 0$, and $D(s, j)$ is finite exactly when $s \leq j \leq k$ for $s \geq 1$. Each globally reachable predecessor set that can extend to a further endpoint is a contiguous interval or the initial singleton.*

The proof is in Electronic Companion Section EC.5.12.

The following pseudocode uses one-based cap indices and zero for the empty deterministic prefix. “Search” means ordinary full-matrix SMAWK on the supplied key. All prior-layer values are known before a search.

Terminal interface. Rows and columns are $1, \dots, k$. At (i, ℓ) return $(1, 0, -\ell)$ if $\ell < i$; otherwise minimize (60) by (61) and return $(0, Q(i, \ell), \ell)$. Search all rows. Reevaluate each winning cell once to store T_i and its continuous optimizer.

Anchored-prefix interface. To form layer $s+1$, use rows $j = s+1, \dots, k$. Columns are $\{1\}$ for $s = 1$, and $s, \dots, k-1$ otherwise. At (j, i) return $(1, 0, i)$ if $i \geq j$, and $(0, P(s, i) + E(i, j), i)$ otherwise. Search and

store values and predecessors. Scan the current reachable endpoints with T_i to update the at-most-budget incumbent; backtrack only its winning codebook.

Deterministic-prefix interface. To form layer s , use rows $j = s, \dots, k$. Columns are $\{0\}$ for $s = 1$, and $s - 1, \dots, k - 1$ otherwise. Return $(1, 0, i)$ if $i \geq j$, and $(0, D(s - 1, i) + C(i + 1, j), i)$ otherwise. Search, store predecessors, and backtrack the solution ending at k .

Lemma E.5 justifies every discarded column. The terminal interface has shrinking feasible suffixes; the program interfaces have expanding feasible prefixes. The completion lemmas include infeasible rows in submatrices created internally by SMAWK, although every original searched row has a feasible entry. Padded comparisons do not evaluate an undefined economic cost.

Theorem E.6 (Contractual corollary of classical matrix search). *For ordered rational inputs satisfying (56) and $1 \leq m \leq k$, both expected-participation and deterministic/pathwise frontiers through budget m , including an optimal codebook for every budget, require $O(mk)$ rational operations and comparisons and $O(mk + k)$ stored rational numbers and indices. The highest randomized level remains continuous. With total input length L and $S = O(k[L + \log(k + 2)])$, a conservative Turing bound is $O(mkS^3)$. Sorting unordered distinct caps adds $O(k \log k)$ exact comparisons.*

The proof is in Electronic Companion Section EC.5.13.

The retained divide-and-conquer implementation is a valid $O(mk \log(k + 1))$ alternative using ordinary row-minimum monotonicity. It is an implementation comparator rather than another headline theorem. $O(k)$ exact-rational oracle/comparison work per layer does not imply uniform wall-time superiority for a particular Python implementation.

F Catalog Approximation Without Promise or Risk Slack

Finite catalogs have two different interpretations. An externally certified menu is a restriction of the feasible technology, for which an exact catalog optimum is the relevant answer. Alternatively the operator may choose a numerical catalog to approximate continuous design. The latter interpretation needs a quantitative comparison to the unrestricted optimum; exactness on the catalog alone does not supply it.

Let $\mathcal{A} \subset [0, 1]$ contain zero and one, and let h be its largest consecutive spacing. Assume f is increasing, concave, and L_f -Lipschitz, and each h_j is convex, nondecreasing, and H_j -Lipschitz on $[0, b_j]$. Let the nonnegative charge ρ be L_ρ -Lipschitz on $[0, 1]$. Define

$$K_m = L_f + \sum_j \pi_j H_j + m L_\rho. \quad (69)$$

For the quadratic model, one can take $L_f = r$ and $H_j = \gamma_j b_j$. Neither the root promise nor the risk tolerance is discretized.

Theorem F.1 (Uniform exactly feasible catalog certificate). *For every $B \in [0, \bar{b}]$, every $\delta \geq 0$, and every $m \geq 1$,*

$$0 \leq \mathcal{V}_m^\delta(B; \rho) - \mathcal{V}_{m, \mathcal{A}}^\delta(B; \rho) \leq K_m h. \quad (70)$$

More strongly, every canonical continuous-book policy has a catalog policy with no more symbols that preserves each branch target exactly, satisfies the same realization ceiling exactly, and loses at most $K_m h$ in net payoff. The conclusion includes the pathwise endpoint $\delta = 0$ and requires no strict feasibility margin.

Proof. Round each original level c downward to $a(c) = \max\{a \in \mathcal{A} : a \leq c\}$ and merge collisions. Fix a branch target t and let d be its old eligible top. Every old eligible level remains eligible after rounding. In particular the new eligible top d' satisfies $d' \geq a(d) \geq d - h$. The old canonical terminal mean is $\mu = \min(t, d)$; coupling its draws to their rounded levels gives a feasible-support lottery of mean $\tilde{\mu} \in [\mu - h, \mu]$ and expected terminal reward at least the old reward minus $L_f h$.

Do not retain those coupled probabilities. Recompute the canonical lottery on the rounded eligible prefix at the same target t . Its terminal mean $\mu' = \min(t, d')$ is at least $\tilde{\mu}$: every rounded old draw is at most d' and its mean is at most t . The new interpolant is increasing, and at mean $\tilde{\mu}$ it dominates the coupled lottery reward. Therefore its reward at μ' also dominates that reward. Moreover $\mu' \geq \mu - h$, so its intermediate tier $t - \mu'$ exceeds the old tier by at most h . The branch operating-payoff loss is at most $(L_f + H_j)h$. Theorem 2.1 ensures exact mean and risk feasibility for this reoptimized lottery, including when the risk bound is binding. All t_j are unchanged, so the root promise remains exactly B .

For each original level, $\rho(a(c)) \leq \rho(c) + L_\rho h$. After merging, the total charge is no larger than the sum of these rounded-slot charges because charges are nonnegative. At most m slots were present. Summing the branch bounds and charge bound gives $K_m h$. Apply the construction to an optimizer, or to a sequence approaching the supremum, and use inclusion of catalog policies in continuous policies for the other inequality. $\square$

One-sided rounding is necessary but not sufficient without the probability repair. For example, let $b = t = 1/2$, $\delta = 1/8$, and old support $(1/4, 5/8)$ with upper probability $2/3$. Rounding the lower level to zero while retaining the probabilities reduces the mean to $5/12$. Compensating by $y = 1/12$ produces a maximum obligation $17/24 > 5/8$, violating the risk limit. Instead, the new canonical lottery on $(0, 5/8)$ uses upper probability $4/5$ and $y = 0$. It preserves the target and the ceiling exactly. This is the distinction between unconstrained quantizer rounding and the contractual transport in Theorem F.1.

Corollary F.2 (Certified approximation for a fixed alphabet budget). *For rational quadratic operating primitives and a nonnegative charge whose rational catalog evaluations and Lipschitz bound are supplied, enumerate every feasible catalog book of size at most m and solve its allocation by (4). With $N = |\mathcal{A}|$, this gives an exact catalog optimum in*

$$O\left(\left[\sum_{s=1}^{\min(m, N)} \binom{N}{s}\right] (km + k \log(k+1))\right) \quad (71)$$

rational arithmetic/evaluation operations. A uniform mesh with denominator $\lceil K_m/\varepsilon \rceil$ (or one when that ceiling is zero) gives additive error at most ε . The algorithm is polynomial in k , K_m/ε , and the bit complexity of the sampled data for fixed m ; it is not claimed to be polynomial in a variable m , nor is this a multiplicative approximation ratio.

Proof. Increasing subsets exhaust the allowed books and the supporting-price algorithm optimizes every inner problem exactly. Take the largest charge-adjusted value and apply Theorem F.1. The mesh includes zero, so every feasible promise is implementable even when it is smaller than the first positive grid point. The bound counts evaluation work; a black-box charge oracle is not assumed to have unit bit complexity unless that is explicitly part of the input model. $\square$

The guarantee is uniform in the two contractual constraints. For any institution pair, subtracting their certified value intervals gives a certified interval for the institutional gap. This permits a global accuracy statement away from saturation without presuming that original cap anchors or Monge order survive. When $B = \bar{b}$, the faster charged path recurrence in Section EC.9 replaces subset enumeration. When the saturated expected problem is uncharged, the continuous Monge algorithm avoids a mesh entirely. Thus the specialized algorithms are faster members of one feasible-policy hierarchy, while (70) connects numerical catalogs to continuous risk-and-charge design.

F.1 The uniform-mesh rate is sharp

The linear bound in Theorem F.1 reflects exact feasibility, not only an insensitive Lipschitz estimate. It cannot be uniformly replaced by a higher-order rate for the declared uniform catalogs.

Proposition F.3 (A fixed-input lower bound). *Take one branch and one symbol, $b = B = 1/2$, $f(c) = c - c^2/2$, $h(y) = y^2/2$, and zero charges. For every risk tolerance and every odd integer $N \geq 3$, the uniform catalog with spacing $h = 1/N$ has loss*

$$\mathcal{V}_1^\delta(B; 0) - \mathcal{V}_{1, \mathcal{A}_N}^\delta(B; 0) = \frac{h}{4} + \frac{h^2}{4}.$$

Thus the worst-case $O(h)$ rate is order-sharp, even for a single fixed input with no opening charges and no randomization.

Proof. With one level, $c \leq B$ and $y = B - c$. Payoff $f(c) - (B - c)^2/2$ is strictly increasing on $[0, B]$. Its continuous maximizer is B , with value $3/8$. For odd N , the largest feasible catalog point is $B - h/2$. Substitution gives the displayed loss. All realizations equal total service B , so risk tolerance plays no role. The sequence of odd N uses the same primitives throughout. $\square$

This is a lower bound for uniform catalogs, not for all adaptive discretizations. Adding the cap $1/2$ removes this particular error. It also does not rule out faster convergence on other fixed instances. The search interval in (51) is therefore useful even when its optimization gap has already vanished: refinement addresses a genuinely different source of error.

References

- Aardal K, van den Berg P, Gijswijt D, Li S (2014) Approximation algorithms for hard capacitated k -facility location problems. Author manuscript, arXiv:1311.4759v4, September 12.
- Aggarwal A, Klawe MM, Moran S, Shor P, Wilber R (1987) Geometric applications of a matrix-searching algorithm. *Algorithmica* 2:195–208. doi:10.1007/BF01840359.
- Chan TM (2021) (Near-)linear-time randomized algorithms for row minima in Monge partial matrices and related problems. *Proceedings of the 2021 ACM–SIAM Symposium on Discrete Algorithms*, 1465–1482. doi:10.1137/1.9781611976465.88.
- Croci M, Fasi M, Higham NJ, Mary T, Mikaitis M (2022) Stochastic rounding: Implementation, error analysis and applications. *Royal Society Open Science* 9:211631. doi:10.1098/rsos.211631.
- Del Pia A, Dey SS, Molinaro M (2017) Mixed-integer quadratic programming is in NP. *Mathematical Programming* 162:225–240. doi:10.1007/s10107-016-1036-0.
- Eppstein D (2005) PADS: Python Algorithms and Data Structures. *SMAWK.py*, offline matrix-search implementation, originally written 2002 and revised 2005. University of California, Irvine. MIT-licensed distribution; source version and excerpt are identified in the code supplement.
- Fattahi A (2025) Utilities’ managed home charging programs for electric vehicles. *Management Science* 71(12):10205–10224. doi:10.1287/mnsc.2023.00850.
- Fu M (2024) A tutorial on quantized feedback control. *IEEE/CAA Journal of Automatica Sinica* 11(1):5–17. doi:10.1109/JAS.2023.123972.
- Grønlund A, Larsen KG, Mathiasen A, Nielsen JS, Schneider S, Song M (2017) Fast exact k -means, k -medians and Bregman divergence clustering in 1D. Preprint, arXiv:1701.07204, version 4.
- Handler GY, Zang I (1980) A dual algorithm for the constrained shortest path problem. *Networks* 10(4):293–309. doi:10.1002/net.3230100403.
- Huangfu Q, Hall JAJ (2018) Parallelizing the dual revised simplex method. *Mathematical Programming Computation* 10(1):119–142. doi:10.1007/s12532-017-0130-5.
- Hübner T, Gupte A, Rebennack S (2025) Spatial branch-and-bound for nonconvex separable piecewise linear optimization. *INFORMS Journal on Computing*, published online June 27. doi:10.1287/ijoc.2024.0755.
- Jourdain B, Pagès G (2021) Optimal dual quantizers of 1D log-concave distributions: Uniqueness and Lloyd like algorithm. *Journal of Approximation Theory* 267:105581. doi:10.1016/j.jat.2021.105581.
- Kaplan H, Mozes S, Nussbaum Y, Sharir M (2017) Submatrix maximum queries in Monge matrices and partial Monge matrices, and their applications. *ACM Transactions on Algorithms* 13(2), Article 26. doi:10.1145/3039873.

- Klawe MM, Kleitman DJ (1990) An almost linear time algorithm for generalized matrix searching. *SIAM Journal on Discrete Mathematics* 3(1):81–97. doi:10.1137/0403009.
- Lawler EL, Wood DE (1966) Branch-and-bound methods: A survey. *Operations Research* 14(4):699–719. doi:10.1287/opre.14.4.699.
- Lemaréchal C (2001) Lagrangian relaxation. Jünger M, Naddef D, eds. *Computational Combinatorial Optimization*, Lecture Notes in Computer Science 2241 (Springer), 112–156. doi:10.1007/3-540-45586-8_4.
- Li Y, Bertsekas D (2025) An error bound for aggregation in approximate dynamic programming. Author manuscript, arXiv:2507.01324v1, July 2.
- Liao W, Wu J (2015) Matrix inequalities for the difference between arithmetic mean and harmonic mean. *arXiv:1501.04823*.
- Nielsen F, Nock R (2014) Optimal interval clustering: Application to Bregman clustering and statistical mixture learning. Author manuscript, arXiv:1403.2485.
- Pagès G, Wilbertz B (2012) Intrinsic stationarity for vector quantization: Foundation of dual quantization. Author manuscript, arXiv:1010.4642v2, March 27. Original preprint 2010.
- Patriksson M, Strömberg C (2015) Algorithms for the continuous nonlinear resource allocation problem—new implementations and numerical studies. Author manuscript, arXiv:1501.07035.
- Schoot Uiterkamp MHH, Gerards MET, Hurink JL (2022) Quadratic nonseparable resource allocation problems with generalized bound constraints. *INFORMS Journal on Optimization* 4(2):215–247. doi:10.1287/ijoo.2021.0065.
- Spear SE, Srivastava S (1987) On repeated moral hazard with discounting. *Review of Economic Studies* 54(4):599–617. doi:10.2307/2297484.
- Stouffer K, Pease M, Tang C, Zimmerman T, Pillitteri V, Lightman S, Hahn A, Saravia S, Sherule A, Thompson M (2023) Guide to operational technology (OT) security. NIST Special Publication 800-82 Rev. 3. doi:10.6028/NIST.SP.800-82r3.
- Thomas J, Worrall T (1988) Self-enforcing wage contracts. *Review of Economic Studies* 55(4):541–554. doi:10.2307/2297404.
- Virtanen P, Gommers R, Oliphant TE, et al. (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods* 17:261–272. doi:10.1038/s41592-019-0686-2.
- Wu X (1991) Optimal quantization by matrix searching. *Journal of Algorithms* 12(4):663–673. doi:10.1016/0196-6774(91)90039-2.
- Yao FF (1980) Efficient dynamic programming using quadrangle inequalities. *Proceedings of the Twelfth Annual ACM Symposium on Theory of Computing*, 429–435. doi:10.1145/800141.804691.

Zhang H (2012) Solving an infinite horizon adverse selection model through finite policy graphs.
Operations Research 60(4):850–864. doi:10.1287/opre.1120.1056.

Zuse Institute Berlin (2026) PySCIPOpt documentation: Non-linear expressions. Official documentation,
accessed September 24.
<https://pyscipopt.readthedocs.io/en/latest/tutorials/expressions.html>.

Table 1: Joint-design regimes and theorem provenance. Each guarantee retains its stated assumptions.

Regime	Guarantee	Dependence	Provenance
Saturated catalog	Exact joint recurrence	Polynomial at saturation	R42; retained companion
Unrestricted price	Exact priced menu	Polynomial per price, not a joint optimum	R43; Appendix B
Fixed individual targets	Exact memory-charge frontier	Polynomial conditional on targets	R45; Theorem 3.1
Individual target boxes	Exact two-sided bound; checked cover	Accuracy search in response types	R46; Theorems 6.1–7.1
Guarded / harmonic groups	Original-space error composition	Accuracy-dependent cap and cost bins	R47–R48; companion
One eligible prefix	Exact original-budget joint optimum	Polynomial in histories, catalog and budget	New; Theorem 4.2
Several eligible prefixes	Exact group oracle; additive joint accuracy	Exponent $E-1$, no dispersion allowance	New; Theorems 8.1–8.2

The full R42–R48 theorem map remains in Table CR.3 of the computational record. Conditional, augmented-budget and original-budget results are not interchanged.

Table 2: All 24 paired original-space intervals. The unchanged tolerance is 10^{-3} and the node allowance is 31.

Case	Histories	Catalog	Budget	Classes	Prior gap	Pooled gap	Nodes	Seconds
00	4	7	1	1	0.0000727	0	1	0.003
01	6	7	3	1	0.0175658	0	1	0.004
02	8	7	2	3	0.001003	0	1	0.025
03	4	7	3	3	0.000022	0.00002196	1	0.024
04	6	7	1	1	0	0	1	0.004
05	8	7	3	1	0.0252872	0	1	0.005
06	4	7	2	3	0	0	1	0.017
07	6	7	3	3	0.0000079	0.00000788	1	0.035
08	8	7	1	1	0.0005123	0	1	0.005
09	4	7	3	1	0.0008874	0	1	0.003
10	6	7	2	3	0.0031844	0	1	0.02
11	8	7	3	3	0.0005434	0.00001723	1	0.041
12	32	9	1	1	0.0000124	0	1	0.025
13	32	9	3	1	0.0264696	0	1	0.024
14	32	9	2	3	0.0027258	0	1	0.135
15	32	9	3	3	0.0004471	0.00013796	5	0.509
16	128	9	1	1	0.0000115	0	1	0.16
17	128	9	3	1	0.0290573	0	1	0.111
18	128	9	2	3	0.0031839	0	1	0.924
19	128	9	3	3	0.0000496	0.00008945	1	0.743
20	256	9	1	1	0.0000041	0	1	0.634
21	256	9	3	1	0.0315955	0	1	0.269
22	256	9	2	3	0.0035516	0	1	2.938
23	256	9	3	3	0	0	1	1.538

Prior gaps are the immutable harmonic-coarsening intervals, not fresh matched-machine measurements. All new intervals meet tolerance; 19 have zero rational width. Case 19 retains a slightly wider interrupted interval. Exact arithmetic and complete lotteries are in the machine-readable certificates.

Table 3: Common-eligibility joint optimization: all 16 intervals are exactly closed in one node.

Histories	Catalog	Exact / cases	Optimization seconds	Checking seconds
16	9	2/2	0.012–0.012	0.009–0.009
16	17	2/2	0.028–0.028	0.020–0.020
64	9	2/2	0.043–0.044	0.030–0.031
64	17	2/2	0.098–0.099	0.072–0.072
256	9	2/2	0.168–0.171	0.121–0.123
256	17	2/2	0.383–0.384	0.287–0.292
1024	9	2/2	0.682–0.685	0.485–0.488
1024	17	2/2	1.530–1.576	1.166–1.191

Each row contains the two declared seeds; time ranges are reported without a statistical confidence claim. The budget is three and the requested tolerance is zero. Caps and service costs vary within the single eligibility class.

Table 4: Eligibility stress at tolerance 10^{-3} . Every row has 12 histories, nine catalog levels and budget three.

Seed	Classes	Exact value	Gap (31)	Gap (127)	Seconds (127)	Complete
49500	6	0.302083	0.0029167	0.0017708	2.825	No
49501	7	0.53564	0.0019627	0.0012862	3.192	No
49502	6	0.459188	0.0022344	0.0009966	3.343	Yes
49503	8	0.030152	0.0017608	0.0015092	1.979	No
49504	5	0.464255	0.0012245	0.0009165	1.431	Yes
49505	6	0.436758	0.009043	0.0059058	3.144	No

Parentheses give node allowances, not claims that every run used all nodes. Two of the twelve tighter requests finish; all twelve requests at 10^{-2} finish. The record includes both tolerances, both budgets, exact references and every interruption. The final column refers to the 127-node allowance.